

第一章 矢量分析

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1、给定三个矢量 $\vec{A}$ 、 $\vec{B}$ 、 $\vec{C}$ 如下:

$$\vec{A} = \vec{e}_x + 2\vec{e}_y - 3\vec{e}_z$$

$$\vec{B} = -4\vec{e}_y + \vec{e}_z$$

$$\vec{C} = 5\vec{e}_x - 2\vec{e}_z$$

求: (1) 单位矢量 $\vec{e}_A$; (2) $|\vec{A} - \vec{B}|$; (3) $\vec{A} \cdot \vec{B}$; (4) $\vec{A}$ 在 $\vec{B}$ 上的分量; (5) $\vec{A} \cdot (\vec{B} \times \vec{C})$ 。

解: (1) $A = |\vec{A}| = [1^2 + 2^2 + 3^2]^{\frac{1}{2}} = \sqrt{14}$

$$\vec{e}_A = \frac{\vec{A}}{A} = \vec{e}_x \cdot \frac{1}{\sqrt{14}} + \vec{e}_y \cdot \frac{2}{\sqrt{14}} + \vec{e}_z \cdot \frac{-3}{\sqrt{14}}$$

$$(2) \vec{A} - \vec{B} = \vec{e}_x + \vec{e}_y + \vec{e}_z \quad (1, 1, 1)$$

$$\therefore |\vec{A} - \vec{B}| = [1^2 + 1^2 + 1^2]^{\frac{1}{2}} = \sqrt{3}$$

$$(3) \vec{A} \cdot \vec{B} = (\vec{e}_x \cdot 2) + (\vec{e}_y \cdot (-4)) + (\vec{e}_z \cdot 1)$$

$$= -8 + 1 = -7$$

$$(4) |\vec{A}| \cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \frac{-7}{(16+1)^{\frac{1}{2}}} = -\frac{7}{\sqrt{17}} = -\frac{7}{\sqrt{17}}$$

$$(5) \vec{B} \times \vec{C} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ 0 & -4 & 1 \\ 5 & 0 & -2 \end{vmatrix} = 8\vec{e}_x + 5\vec{e}_y - (-20\vec{e}_z)$$

$$\therefore \vec{A} \cdot (\vec{B} \times \vec{C}) = (1, 2, -3) \cdot (8, 5, 20)$$

$$= 8 + 10 - 60 = -42$$

4、给定两矢量 $\vec{A} = 2\vec{e}_x + 3\vec{e}_y - 4\vec{e}_z$ 和 $\vec{B} = 4\vec{e}_x - 5\vec{e}_y + 6\vec{e}_z$ ，求它们之间的夹角和 $\vec{A}$ 在 $\vec{B}$ 上的分量。

解: $\cos\theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \cdot |\vec{B}|} = \frac{8 - 15 - 24}{(\sqrt{2^2 + 3^2 + 4^2})^{\frac{1}{2}} (\sqrt{4^2 + 5^2 + 6^2})^{\frac{1}{2}}} = \frac{-31}{\sqrt{29} \times \sqrt{77}} = -0.656$

$\therefore \theta = \arccos(-0.656) = 131^\circ$

公式对

$\therefore |\vec{A}| \cos\theta = -0.656 |(\vec{e}_x \cdot 2 + \vec{e}_y \cdot 3 - \vec{e}_z \cdot 4)| = -3.532$

$\therefore \vec{A}$ 在 $\vec{B}$ 上的分量是 $-3.532 \vec{B} = \vec{e}_x(-14.13) + \vec{e}_y 17.66 + \vec{e}_z(-21.19)$

$\vec{A} \cdot \vec{e}_B = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \dots$

圆柱坐标与直角坐标转换

3、在圆柱坐标中，一点的位置由 $(4, \frac{2\pi}{3}, 3)$ 定出，求该点在：(1) 直角坐标中的坐标；(2) 球坐标中的坐标。

解: $\rho = 4, \theta = \frac{2\pi}{3}, z = 3$

(1) 转直角坐标 $x = \rho \cos\theta = 4 \times \frac{1}{2} = 2 \times (-1) = -2$

$y = \rho \sin\theta = 4 \times \frac{\sqrt{3}}{2} = 2\sqrt{3}$

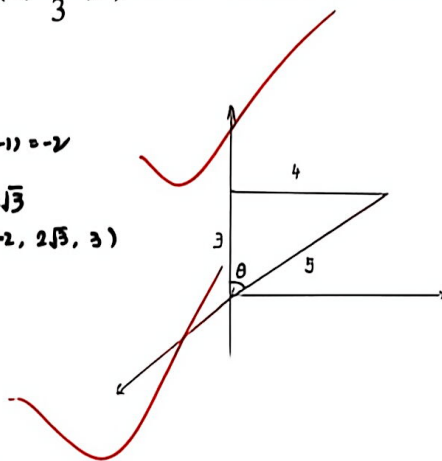
$z = 3$ ，坐标 $(-2, 2\sqrt{3}, 3)$

(2) 转球坐标 $r = (3^2 + 4^2)^{\frac{1}{2}} = 5$

$\theta = \arctan(\frac{4}{3}) = 53^\circ$

$\phi = \frac{2\pi}{3} = 120^\circ$

$\therefore$ 坐标 $(5, 53^\circ, 120^\circ)$



(3分)

1、如图，求

解：

4、一球面 S 的半径为 5，球心在原点上，计算： $\oint_S (\vec{e}_r, 2 \sin \theta) \cdot d\vec{S}$ 的值。有效的积分计算解： $d\vec{S} = \vec{e}_r dS = \vec{e}_r r^2 \sin \theta d\theta d\phi$ ，其中 $r=5$

$$\oint_S (\vec{e}_r, 2 \sin \theta) \cdot d\vec{S} = \oint_S 2 \times 5 \sin \theta d\theta d\phi = \int_0^{2\pi} \int_0^\pi 25 (1 - \cos 2\theta) d\theta d\phi = 50\pi$$

有效面积元 $d\vec{S}$ 用矢量展开 $d\vec{S} = \vec{e}_r dS$

$$\oint_S \vec{e}_r 2 \sin \theta \vec{e}_r dS = \int_0^{2\pi} \int_0^\pi 2 \sin \theta \times 5^2 \sin \theta d\theta d\phi$$

$$= \int_0^{2\pi} \frac{\pi}{2} \times 2 \times 25 d\phi = 50\pi$$

2、求

解

* 5、在由 $r=5$ 、 $z=0$ 和 $z=4$ 围成的圆柱形区域，对矢量 $\vec{A} = r^2 \vec{e}_r + 2z \vec{e}_z$ 验证散度定理。

$$\text{解：} \int_V \nabla \cdot \vec{F} dV = \oint_S \vec{F} \cdot d\vec{S}, \quad \nabla \cdot \vec{A} = \frac{1}{r} \frac{\partial}{\partial r} (r \cdot r^2) + \frac{\partial 2z}{\partial z} = \frac{1}{r} \cdot 3r^2 + 2$$

$$\text{其中 } \int_V \nabla \cdot \vec{A} dV = \int_V \left(\frac{\partial A_r}{\partial r} + \frac{\partial A_z}{\partial z} \right) dV = \int_0^4 \int_0^{2\pi} \int_0^5 (3r + 2) r dr d\phi dz = \int_0^4 (r^2 + r^2) \Big|_0^5 d\phi dz$$

$$= 150 \times 4 \times 2\pi = 1200\pi$$

$$\oint_S \vec{A} \cdot d\vec{S} = \int_{z=0} 2z \vec{e}_z \cdot (-\vec{e}_n) dS + \int_{z=4} 2 \times 4 \vec{e}_z \cdot \vec{e}_n dS + \int_S r^2 \vec{e}_r \cdot \vec{e}_r dS$$

$$= 2 \times 0 \times (-\pi r^2) + 2 \times 4 \times (\pi r^2) + \int_0^4 \int_0^{2\pi} r^2 \cdot r d\phi dz$$

$$= 8 \times 25\pi + 2\pi \times 5^3 \times 4 = 1200\pi$$

$$\text{故 } \int_V \nabla \cdot \vec{F} dV = \oint_S \vec{A} \cdot d\vec{S}, \text{ 可验证散度定理}$$

6、求矢量 $\vec{A} = x\vec{e}_x + x^2\vec{e}_y + y^2z\vec{e}_z$ 沿 xy 平面上一个边长为 2 的正方形回路的线积分，此正方形的两边分别与 x 和 y 轴相重合。再求 $\nabla \times \vec{A}$ 对此回路所包围的曲面积分，验证斯托克斯定理。

解: $\int_C \vec{A} \cdot d\vec{l} = \int_0^2 x dx + \int_0^2 x^2 dy \Big|_{x=2} + \int_2^0 x^2 dy \Big|_{x=0} + \int_2^0 x dx$
 $= -\int_0^2 0 dy + \int_0^2 4 dy = +8$

$$\nabla \times \vec{A} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & x^2 & y^2z \end{vmatrix} = (2yz - 0)\vec{e}_x + (0)\vec{e}_y + (2x)\vec{e}_z$$

$\therefore \int_{\text{面}} \nabla \times \vec{A} \cdot d\vec{S} = \int_0^2 \int_0^2 2x \cdot dx dy = 2 \cdot x^2 \Big|_0^2 = 8$, 被积展开, 验证成立;
 积分展开, 验证成立;

$\therefore \int_C \vec{A} \cdot d\vec{l} = \int_S \nabla \times \vec{A} \cdot d\vec{S}$, Stokes 定理验证

7、求标量函数 $\psi = x^2yz$ 的梯度及 ψ 在一个指定方向的方向导数，此方向由单位矢量

$\vec{e}_x \frac{3}{\sqrt{50}} + \vec{e}_y \frac{4}{\sqrt{50}} + \vec{e}_z \frac{5}{\sqrt{50}}$ 定出：求 (2,3,1) 点的方向导数值。

解: $\nabla \psi = \vec{e}_x \frac{\partial \psi}{\partial x} + \vec{e}_y \frac{\partial \psi}{\partial y} + \vec{e}_z \frac{\partial \psi}{\partial z}$
 $= \vec{e}_x 2xyz + \vec{e}_y x^2z + \vec{e}_z x^2y$

方向上的方向余弦为 $\cos \alpha = \frac{3}{\sqrt{50}} / 1 = \frac{3}{\sqrt{50}}$

$\cos \beta = \frac{4}{\sqrt{50}} / 1 = \frac{4}{\sqrt{50}}$

$\cos \gamma = \frac{5}{\sqrt{50}} / 1 = \frac{5}{\sqrt{50}}$

$\therefore \frac{\partial \psi}{\partial l} = 2xyz \cos \alpha + x^2z \cos \beta + x^2y \cos \gamma; (x,y,z) = P(2,3,1)$

$\therefore \left(\frac{\partial \psi}{\partial l} \right)_P = \frac{18}{\sqrt{50}} + \frac{8}{\sqrt{50}} + \frac{6}{\sqrt{50}} = \frac{56}{\sqrt{50}}$

$\nabla \psi \cdot \vec{e}_l =$

$\frac{\partial \psi}{\partial l} = \nabla \psi \cdot \vec{e}_l = \left(\vec{e}_x \cdot \frac{3}{\sqrt{50}} \right) \cdot \left(\vec{e}_x \cdot 2xyz \right) + \left(\vec{e}_y \cdot \frac{4}{\sqrt{50}} \right) \cdot \left(\vec{e}_y \cdot x^2z \right) + \left(\vec{e}_z \cdot \frac{5}{\sqrt{50}} \right) \cdot \left(\vec{e}_z \cdot x^2y \right)$
 $= \frac{3}{\sqrt{50}} \times 12 + \frac{4}{\sqrt{50}} \times 4 + \frac{5}{\sqrt{50}} \times 12 = \frac{112}{\sqrt{50}}$, 正确.

已知矢量场 $\vec{A} = (axz + x^2) \vec{e}_x + (by + xy^2) \vec{e}_y + (z - z^2 + cxz - 2xyz) \vec{e}_z$ ，试确定 a 、 b 、 c ，

使得 $\vec{A}$ 成为一无源场。无源场上散度置零，并公式展开计算

解：无源场散度 $\nabla \cdot \vec{A} = 0$

$$\text{故 } \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 0$$

$$(axz + 2x) + (b + 2xy) + (z - 2z + cx - 2xy) = 0 \quad \text{恒成立}$$

对 x, y, z 而言 解得

$$\begin{cases} 2x + cx = 0 \\ axz - 2z = 0 \\ b + 1 = 0 \end{cases} \quad \begin{cases} a = 2 \\ b = -1 \\ c = -2 \end{cases}$$

作业注意事项：

- 1、请用 A4 纸正、反面双面打印；
- 2、作业过程手写，注意解题过程；
- 3、订书机安装位置统一在左上角；
- 4、作业批改下发后请妥善保管，部分作业需要备学校抽查。

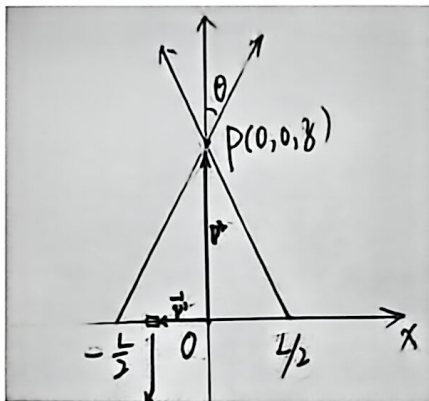
第2章 电磁场的基本规律

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(3解即可)

1、如图, 求一有限长均匀带电直线段 L , 线电荷密度为 ρ_l , 求距离直线段中点为 z 处的电场强度 $\vec{E}$ 。

解:

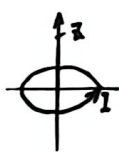


x处dq 题1图

$$\begin{aligned}
 d\vec{E} &= \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{|\vec{r}-\vec{r}'|^3} (\vec{r}-\vec{r}') \quad \text{且具有对称性} \\
 dq &= \rho_l dx \\
 \text{故 } \vec{E} &= \int d\vec{E} = \frac{1}{4\pi\epsilon_0} \int_{-L/2}^{L/2} \frac{\rho_l}{\frac{1}{2}(x^2+z^2)^{3/2}} (x\vec{e}_x + z\vec{e}_z) dx \\
 &= \frac{\rho_l}{4\pi\epsilon_0} \int_{-L/2}^{L/2} (x^2+z^2)^{-3/2} \cdot z\vec{e}_z dx \\
 \text{注意到 } \frac{x}{z} &= \tan\theta, \text{ 则 } x^2+z^2 = z^2(1+\tan^2\theta) = z^2\sec^2\theta, z' = z\sec^2\theta \cdot d\theta \\
 \int_{-L/2}^{L/2} (x^2+z^2)^{-3/2} \cdot z\vec{e}_z dx &= \int_{-\theta}^{\theta} [z^{-3}\sec^{-3}\theta] \cdot [z\sec^2\theta \cdot d\theta] \cdot z\vec{e}_z \\
 &= \int_{-\theta}^{\theta} z^{-1} \cos\theta d\theta = z^{-1} \sin\theta \\
 \therefore \vec{E} &= \frac{\rho_l L}{z\sqrt{z^2+\frac{L^2}{4}}} \cdot \frac{\rho_l}{4\pi\epsilon_0} = \frac{\rho_l \cdot L}{2\pi\epsilon_0 \cdot z\sqrt{z^2+\frac{L^2}{4}}} \cdot \vec{e}_z
 \end{aligned}$$

2、求电流为 I , 半径为 a 的小圆环在远离圆环中心垂直方向 z 处的磁感应强度 $\vec{B}$ 。

解

源点 $\vec{r}'(a, 0, 0)$ 圆柱坐标 (ρ, ϕ, z) 场点 $\vec{r}(0, 0, z)$

$$\text{故 } \vec{e}_{\vec{r}-\vec{r}'} = \frac{z\vec{e}_z - a\vec{e}_\rho}{\sqrt{z^2+a^2}}, \quad \vec{R} = z\vec{e}_z - a\vec{e}_\rho$$

$$\text{由 } d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{I d\vec{l} \times \vec{R}}{R^3}, \text{ 其中 } d\vec{l} \times \vec{R} = (a d\phi \vec{e}_\phi) \times (z\vec{e}_z - a\vec{e}_\rho)$$

$$= a \cdot (z\vec{e}_\phi + a\vec{e}_z) d\phi$$

$$\therefore \vec{B} = \frac{\mu_0 I a}{4\pi} \int_0^{2\pi} d\phi \cdot (z\vec{e}_\phi + a\vec{e}_z) \cdot \frac{1}{(z^2+a^2)^{3/2}}$$

$$= \frac{\mu_0 I a}{4\pi} \cdot \frac{2\pi a}{(z^2+a^2)^{3/2}} \vec{e}_z$$

$$= \frac{\mu_0 I a^2}{2(z^2+a^2)^{3/2}} \vec{e}_z$$

明白一个抽象事，计算球外分布时 $\int_V \rho dV = \int_0^a \frac{Q}{4\pi a^3} \cdot d(\frac{4}{3}\pi r^3) = \int_0^a \frac{3Q}{4\pi a^3} \cdot 4\pi r^2 dr$

2025-2026 学年第二学期《电磁场与电磁波》作业

✓、半径为 a 的球形带电体，电荷总量 Q 均匀分布在球体内。

求：(1) $\vec{E}(\vec{r})$ ；(2) $\nabla \cdot \vec{E}(\vec{r})$ ；(3) $\nabla \times \vec{E}(\vec{r})$ ；

解：当 $0 < r < a$ 时 (1) (2) (3) 由下式得出

$$\nabla \cdot \vec{E}(\vec{r}) = \rho / \epsilon_0, \text{ 其中 } \rho = \frac{Q}{\frac{4}{3}\pi a^3} \cdot \frac{4}{3}\pi r^3 = \frac{r^3}{a^3} \cdot Q \cdot \frac{1}{\frac{4}{3}\pi r^3} = \frac{3Q}{4\pi a^3}$$

$$\therefore \nabla \cdot \vec{E}(\vec{r}) = \frac{3Q}{4\pi a^3} \cdot \frac{1}{\epsilon_0}, \quad \nabla \times \vec{E}(\vec{r}) = 0$$

转化成微分形式 $\vec{E}(\vec{r}) \cdot d\vec{S}$ 。

当 $r \geq a$ 时 (1) (2) (3) 由下式得出

$$\nabla \cdot \vec{E}(\vec{r}) = \frac{Q}{\frac{4}{3}\pi a^3} \cdot \frac{1}{\epsilon_0} = \frac{3Q}{4\pi a^3} \cdot \frac{1}{\epsilon_0}, \quad \nabla \times \vec{E}(\vec{r}) = 0$$

$$\int_V \nabla \cdot \vec{E} dV = \int_V \frac{\rho}{\epsilon_0} dV = \int \frac{\rho}{\epsilon_0} dV, \text{ 于是可以计算出 } \vec{E}(\vec{r}) \cdot 4\pi r^2 = \frac{Q}{\epsilon_0} \cdot \frac{r^3}{a^3}, \text{ 即 } \vec{E}(\vec{r}) = \frac{Q}{\epsilon_0} \cdot \frac{r}{4\pi a^3} \vec{e}_r (r \geq a)$$

考虑

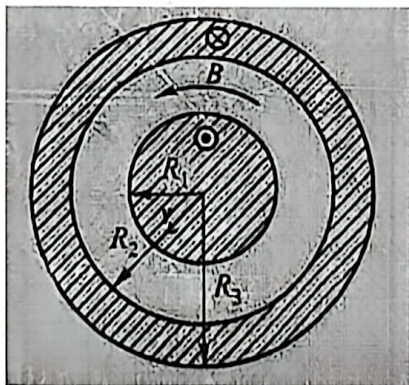
第2题图

$$\nabla \cdot \vec{E}(\vec{r}) = \frac{\rho}{\epsilon_0} = \frac{Q}{\frac{4}{3}\pi a^3 \epsilon_0} = \frac{3Q}{4\pi a^3} \cdot \frac{1}{\epsilon_0}$$

$$\text{考虑 } r \geq a \text{ 的情况, } \int \frac{\rho}{\epsilon_0} dV = \int \rho dV / \epsilon_0 = \frac{Q}{\epsilon_0}, \text{ 即 } \vec{E}(\vec{r}) = \frac{1}{4\pi r^2} \cdot \frac{Q}{\epsilon_0} \vec{e}_r, [\nabla \cdot \vec{E}(\vec{r}) = \frac{\rho}{\epsilon_0} = \frac{3Q}{4\pi a^3} \cdot \frac{1}{\epsilon_0}]$$

4、如图，一无限长同轴电缆芯线通有均匀分布的电流 I ，外导体通有均匀等量反向电流，求各区域的磁感应强度。

解：



题4图

1° $0 \leq r \leq R_1$

考虑 $I_1 = \frac{\pi r^2}{\pi R_1^2} \cdot I$, 那么

$$\oint_C \vec{B}_1 \cdot d\vec{l} = \mu_0 I_1, \text{ 在柱坐标系下展开}$$

$$\vec{B}_1 \cdot \vec{e}_\phi \cdot 2\pi r = \mu_0 \cdot \frac{r^2}{R_1^2} \cdot I$$

$$\therefore \vec{B}_1 = \frac{\mu_0 I}{2\pi R_1^2} \vec{e}_\phi$$

2° $R_1 \leq r \leq R_2$

$I_2 = I$, 那么

$$\oint_C \vec{B}_2 \cdot d\vec{l} = \mu_0 I, \text{ 展开}$$

$$\vec{B}_2 \cdot \vec{e}_\phi \cdot 2\pi r = \mu_0 I$$

$$\therefore \vec{B}_2 = \frac{\mu_0 I}{2\pi r} \vec{e}_\phi$$

3° $R_2 \leq r \leq R_3$ $I_3 = I - \left(\frac{\pi r^2 - \pi R_2^2}{\pi R_3^2 - \pi R_2^2} \right) I$, 那么

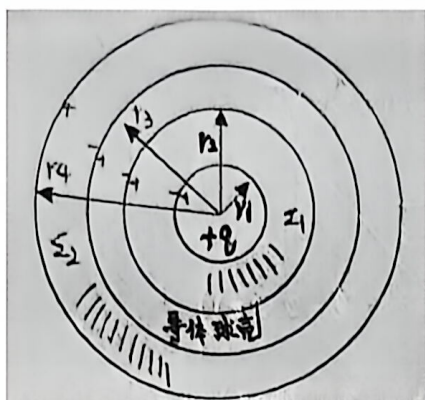
$$\oint_C \vec{B}_3 \cdot d\vec{l} = \mu_0 I_3, \text{ 展开}$$

$$\vec{B}_3 \cdot \vec{e}_\phi \cdot 2\pi r = \mu_0 \cdot \left(1 - \frac{\pi r^2 - \pi R_2^2}{\pi R_3^2 - \pi R_2^2} \right) I$$

$$\therefore \vec{B}_3 = \vec{e}_\phi \cdot \frac{\mu_0}{2\pi r} \left(\frac{\pi R_3^2 - \pi r^2}{\pi R_3^2 - \pi R_2^2} \right) \cdot I$$

4° $r > R_3$. $I_4 = 0$. 故 $\vec{B}_4 = \vec{0}$.

5. 已知半径为 r_1 的导体球携带的正电量为 q ，该导体球被内半径为 r_2 的导体球壳所包围，球与球壳之间填充介质，其介电常数为 ϵ_1 ，球壳的外半径为 r_3 ，球壳的外表面敷有一层介质，该层介质的外半径为 r_4 ，介电常数为 ϵ_2 ，外部区域为真空。试求：(1) 各区域中的电场强度；(2) 求各表面上的自由电荷。



题 5 图

$$\nabla \cdot \vec{D} = \rho$$

$$\oint \vec{D} \cdot d\vec{S} = \int_V \rho dV = q.$$

(1) $0 \leq r < r_1$ 在导体球内, $q=0$, $E=0$

$r_1 \leq r < r_2$ $\oint \vec{D}_2 \cdot d\vec{S} = \int_V \rho dV = +q$, 那么

$$\therefore \vec{D}_2 = \frac{q}{4\pi r^2} \vec{e}_r; \text{ 由 } \vec{D}_2 = \epsilon_0 \epsilon_1 \vec{E}_2 = \epsilon_1 \vec{E}_2$$

$$\therefore \vec{E}_2 = \frac{q}{4\pi \epsilon_0 \epsilon_1 r^2} \vec{e}_r$$

$r_2 \leq r < r_3$ 在导体球壳内, $E_3=0$

$r_3 \leq r < r_4$ $\oint \vec{D}_4 \cdot d\vec{S} = \int_V \rho dV = +q$, 那么

$$\vec{D}_4 = \frac{q}{4\pi r^2} \vec{e}_r, \text{ 由 } \vec{D}_4 = \epsilon_0 \epsilon_2 \vec{E}_4 = \epsilon_2 \vec{E}_4$$

$$\therefore \vec{E}_4 = \frac{q}{4\pi \epsilon_0 \epsilon_2 r^2} \vec{e}_r$$

$r_4 \leq r$ $\oint \vec{D}_5 \cdot d\vec{S} = \int_V \rho dV = +q$, 那么

$$\vec{D}_5 = \frac{q}{4\pi r^2} \vec{e}_r, \text{ 由 } \vec{D}_5 = \epsilon_0 \vec{E}_5$$

$$\therefore \vec{E}_5 = \frac{q}{4\pi \epsilon_0 r^2} \vec{e}_r$$

(2) 导体球表面 (r_1) $+q$.

导体球壳内表面 (r_2) $-q$.

导体球壳外表面 (r_3) $+q$.

介质层 ϵ_2 外表面 (r_4) 0 .

6、在媒质 $\sigma=0$, $\varepsilon=4\varepsilon_0$ 和 $\mu=5\mu_0$ 中, 位移电流密度为 $2\cos(\omega t-5z)\vec{e}_x$ ($2A/m^2$)。(1) 根据位移电流, 求 $\vec{D}$ 和 $\vec{E}$; (2) 用法拉第定律的微分形式和对时间积分求 $\vec{B}$ 和 $\vec{H}$; (3) 利用安培环路定律的微分形式求位移

密度 $\vec{J}_d$, 并求 ω 。 ($\sqrt{\frac{\mu_0}{\varepsilon_0}}=120\pi\Omega, \frac{1}{\sqrt{\mu_0\varepsilon_0}}=3\times 10^8 m/s$)

解: (1) $\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$, 则 $\vec{D} = \int \vec{J}_d dt = \int 2\cos(\omega t-5z)\vec{e}_x dt = \frac{2}{\omega} \sin(\omega t-5z)\vec{e}_x$

$$\vec{D} = 2 \times 10^{-6} \sin(\omega t-5z) \vec{e}_x$$

~~$$\vec{E} = \frac{\vec{D}}{\varepsilon_0 \varepsilon_r} = \frac{2 \times 10^{-6}}{4 \times 8.85 \times 10^{-12}} \sin(\omega t-5z) \vec{e}_x$$~~

$$\vec{D} = \vec{E} \cdot (\varepsilon_0 \varepsilon_r) \quad \vec{E} = \frac{5 \times 10^{-7}}{\omega \varepsilon_0} \sin(\omega t-5z) \vec{e}_x$$

(2) $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

$$\therefore \vec{B} = \int -\nabla \times \vec{E} dt = -\int \frac{\partial}{\partial z} \left[\frac{5 \times 10^{-7}}{\omega \varepsilon_0} \sin(\omega t-5z) \right] dt \vec{e}_y$$

$$= \int \frac{2.5 \times 10^{-6}}{\omega \varepsilon_0} \cos(\omega t-5z) \vec{e}_y dt = \frac{2.5 \times 10^{-6}}{\omega^2 \varepsilon_0} \sin(\omega t-5z) \vec{e}_y$$

$$\therefore \vec{H} = \vec{B} / \mu_0 \mu_r = \frac{2.5 \times 10^{-6}}{\varepsilon_0 \omega^2 \mu_0 \mu_r} \sin(\omega t-5z) \vec{e}_y = \frac{5 \times 10^{-7} \cdot \sin(\omega t-5z)}{\omega^2 \varepsilon_0 \mu_0} \vec{e}_y$$

(3) $\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$, 其中 $\nabla \times \vec{H}$

$$= \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & \frac{2.5 \times 10^{-6}}{\omega^2 \mu_0 \cdot 5} \sin(\omega t-5z) & 0 \end{vmatrix} = -\frac{\partial}{\partial z} \left[\frac{5 \times 10^{-7}}{\omega^2 \mu_0 \cdot \varepsilon_0} \sin(\omega t-5z) \right] \vec{e}_x$$

$$= \frac{2.5 \times 10^{-6}}{\omega^2 \mu_0 \varepsilon_0} \cdot \cos(\omega t-5z) \vec{e}_x$$

由 (3) 式与 $\vec{J}_d = 2\cos(\omega t-5z)\vec{e}_x$ 可知: $\frac{2.5 \times 10^{-6}}{\omega^2 \mu_0 \varepsilon_0} = 2 \times 10^{-6}$

$$\therefore \omega = 3.35 \times 10^8 \text{ rad/s}$$

7、同轴线的内导体半径 $a=1\text{mm}$, 外导体的内半径 $b=4\text{mm}$, 内外导体间为空气, 如图所示。假设内、外导体间的

电场强度为 $\vec{E} = \vec{e}_\rho \frac{100}{\rho} \cos(10^8 t - kz) \text{ V/m}$

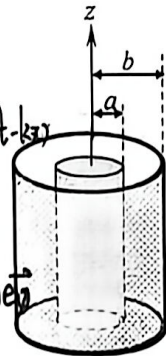
(1) 求与 $\vec{E}$ 相伴的 $\vec{H}$; (2) 确定 k 的值; (3) 求内导体表面的电流密度;

(4) 求沿轴线 $0 \leq z \leq 1\text{m}$ 区域内的位移电流。

解: $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$, 则 $\nabla \times \vec{E} = \frac{1}{\rho} \begin{vmatrix} \vec{e}_\rho & \vec{e}_\phi & \vec{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ \frac{100}{\rho} \cos(10^8 t - kz) & 0 & 0 \end{vmatrix} = \left(\frac{1}{\rho} \cdot \frac{100}{\rho} \cdot (-k \sin(10^8 t - kz)) \right) \vec{e}_\phi = -\frac{100k}{\rho^2} \sin(10^8 t - kz) \vec{e}_\phi$

$$\therefore \vec{B} = -\vec{e}_\phi \cdot \frac{100k}{\rho} \int \sin(10^8 t - kz) dz = -\frac{100k}{\rho} \times \frac{1}{10^8} \cos(10^8 t - kz) \vec{e}_\phi = -\frac{k}{\rho} \times 10^{-6} \cos(10^8 t - kz) \vec{e}_\phi$$

$$\therefore \vec{H} = \vec{B} / \mu_0 \mu_r = \frac{\vec{B}}{\mu_0} = \frac{k}{\rho \mu_0} \times 10^{-6} \cos(10^8 t - kz) \vec{e}_\phi$$



题 8 图

(2) $\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$, 那么

$$\nabla \times \vec{H} = \frac{1}{\rho} \begin{vmatrix} \vec{e}_\rho & \vec{e}_\phi & \vec{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ 0 & \frac{k}{\rho \mu_0} \times 10^{-6} \cos(10^8 t - kz) & 0 \end{vmatrix} = \left[-\frac{\partial}{\partial \rho} \left(\frac{k}{\rho \mu_0} \times 10^{-6} \cos(10^8 t - kz) \right) \vec{e}_\phi + \frac{\partial}{\partial \rho} \left(\frac{k}{\rho \mu_0} \times 10^{-6} \cos(10^8 t - kz) \right) \vec{e}_z \right] \cdot \frac{1}{\rho}$$

$$\therefore \frac{\partial \vec{D}}{\partial t} = -\frac{k^2}{\rho \mu_0} \times 10^{-6} \times \sin(10^8 t - kz) \vec{e}_\phi \Rightarrow \vec{D} = -\int \frac{k^2}{\rho \mu_0} \times 10^{-6} \times \sin(10^8 t - kz) \vec{e}_\phi dt$$

$$\vec{D} = \frac{k^2}{\rho \mu_0} \times 10^{-6} \times \cos(10^8 t - kz) \times \frac{1}{10^8}$$

$$\begin{cases} \vec{D} = \frac{k^2 \times 10^{-14}}{\rho \mu_0} \times (\cos 10^8 t - kz) \vec{e}_\phi \\ \vec{E} = \frac{100}{\rho} \cos(10^8 t - kz) \vec{e}_\rho \end{cases} \Rightarrow k = \frac{1}{3} \text{ rad/m}$$

(3) 内导体表面, 由边界条件

$$\vec{e}_n \times \vec{H} = \vec{J}_s, \text{ 其中 } \vec{H} = \frac{1}{3\rho \mu_0} \times 10^{-6} \cos(10^8 t - \frac{1}{3}z) \vec{e}_\phi$$

$$\therefore \vec{J}_s = \vec{e}_n \times \left(\frac{1}{3\rho \mu_0} \times 10^{-6} \cos(10^8 t - \frac{1}{3}z) \right) \vec{e}_\phi, \text{ 其中 } a = 1 \times 10^{-3} \text{ m}$$

$$= 265.26 \cos(10^8 t - \frac{1}{3}z) \vec{e}_z$$

(4) 位移电流密度 $\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon_r \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \epsilon_0 \cdot \vec{e}_\rho \cdot \frac{100}{\rho} (-\sin(10^8 t - kz)) \times 10^8$

$$= -\frac{1}{\rho} (8.85 \times 10^{-2}) \sin(10^8 t - kz) \vec{e}_\rho$$

$$Id = \int_S \vec{J}_d \cdot d\vec{S} = \int_0^l -\frac{1}{\rho} \cdot 2\pi\rho \cdot (\vec{e}_\rho)^T \cdot (8.85 \times 10^{-2}) \sin(10^8 t - kz) dz, l$$

$$= \int_0^1 -2\pi \times (8.85 \times 10^{-2}) \sin(10^8 t - \frac{1}{3}z) dz$$

$$= -0.55 \left. \frac{\cos(10^8 t - \frac{1}{3}z)}{-\frac{1}{3}} \right|_0^1 = -0.55 \left[\cos(10^8 t - \frac{1}{3}) - \cos(10^8 t) \right]$$

$$= -0.55 \left[\cos(10^8 t - \frac{1}{3}) - \cos(10^8 t) \right];$$

8、媒质 1 的电参数为 $\epsilon_1 = 5\epsilon_0$ 、 $\mu_1 = 3\mu_0$ 、 $\sigma_1 = 0$ ，媒质 2 可视为理想导体 ($\sigma_2 = \infty$)。设 $y=0$ 为理想导体表面， $y>0$ 的区域 (媒质 1) 内的电场强度

$$\vec{E} = \vec{e}_y 20 \cos(2 \times 10^8 t - 2.58z) \quad \text{V/m}$$

试计算 $t=6\text{ns}$ 时：(1) 点 $P(2,0,0.3)$ 处的面电荷密度 ρ_s ；(2) 点 P 处的 $\vec{H}$ ；(3) 点 P 处的面电流密度 $\vec{J}_s$ 。

$$1) \vec{E} = \vec{e}_y 20 \cos(2 \times 10^8 \times 6 \times 10^{-9} - 2.58 \times 0.3) = 14.32 \quad \text{V/m}$$

$$\rho_s = \vec{E} \cdot \vec{D}_1 = \vec{E} \cdot \vec{E} \cdot \epsilon_0 = 1.27 \times 10^{-10} \quad \text{C/m}^2$$

$$2) \text{由 } \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \text{ 其中 } \vec{E} = 20 \cos(2 \times 10^8 t - 2.58z) \vec{e}_y$$

$$\text{其中 } \nabla \times \vec{E} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 20 \cos(2 \times 10^8 t - 2.58z) & 0 \end{vmatrix} = -\frac{\partial}{\partial z} \vec{e}_x (20 \cos(2 \times 10^8 t - 2.58z))$$

$$= -51.6 \vec{e}_x \sin(2 \times 10^8 t - 2.58z)$$

$$= -\frac{\partial \vec{B}}{\partial t}$$

$$\therefore \vec{B} = \int 51.6 \vec{e}_x \sin(2 \times 10^8 t - 2.58z) dt = -\frac{51.6}{2 \times 10^8} \cos(2 \times 10^8 t - 2.58z) \vec{e}_x$$

$$\therefore \vec{H} = \vec{B} / \mu_0 \mu_1 = -0.068 \cos(2 \times 10^8 t - 2.58z) \vec{e}_x \quad \text{即有 } \vec{H} = -0.047 \vec{e}_x$$

$$3) \vec{J}_s = \vec{H} \times \vec{e}_n = \text{边界条件 } \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$$

$$\text{则 } \vec{J}_s = \vec{e}_y \times (-0.047 \vec{e}_x) = -0.047 \vec{e}_z$$

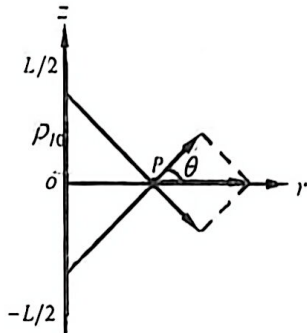
$$0.0623$$

第三章 静态场及其边值问题的解

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1、长度为 L 的细导线带有均匀电荷，其电荷线密度为 ρ_{l0} 。(1) 计算线电荷平分面上任意点的电位 φ ；(2) 利用直接积分法计算线电荷平分面上任意点的电场 $\vec{E}$ ，并用 $\vec{E} = -\nabla\varphi$ 核对。

解:



题1图

$$(1) \text{ 点电荷 } q_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{R}$$

$$\begin{aligned} \text{于是 } \varphi(r) &= \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{1}{4\pi\epsilon_0} \cdot \frac{\rho_{l0} dz}{\sqrt{z^2 + r^2}} = \frac{\rho_{l0}}{4\pi\epsilon_0} \cdot \ln(z + \sqrt{z^2 + r^2}) \Big|_{-\frac{L}{2}}^{\frac{L}{2}} \\ &= \frac{\rho_{l0}}{4\pi\epsilon_0} \cdot \ln\left(\frac{\frac{L}{2} + \sqrt{\frac{L^2}{4} + r^2}}{-\frac{L}{2} + \sqrt{\frac{L^2}{4} + r^2}}\right) = \frac{\rho_{l0}}{4\pi\epsilon_0} \cdot \ln\left(\frac{1 + \sqrt{1 + 4r^2}}{-1 + \sqrt{1 + 4r^2}}\right) \end{aligned}$$

$$(2) \text{ 点电荷 } d\vec{E} = \frac{\rho_{l0} dz}{4\pi\epsilon_0 (z^2 + r^2)^{\frac{3}{2}}} (-z\vec{e}_z + r\vec{e}_r)$$

$$\begin{aligned} \therefore \vec{E}(r) &= \frac{\rho_{l0} \cdot r\vec{e}_r}{4\pi\epsilon_0} \cdot \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{dz}{(z^2 + r^2)^{\frac{3}{2}}} \\ &= \frac{\rho_{l0} \cdot r\vec{e}_r}{4\pi\epsilon_0} \cdot \frac{z}{r^2 \sqrt{z^2 + r^2}} \Big|_{-\frac{L}{2}}^{\frac{L}{2}} \\ &= \frac{\rho_{l0}}{4\pi\epsilon_0} \cdot \frac{L}{r \sqrt{\frac{L^2}{4} + r^2}} \cdot \vec{e}_r \end{aligned}$$

$$\text{检验 } \vec{E} = -\nabla(\varphi) = -\nabla\varphi(r) = -\frac{\rho_{l0}}{4\pi\epsilon_0} \cdot \frac{\partial}{\partial r} \left(\ln\left(\frac{1 + \sqrt{1 + 4r^2}}{-1 + \sqrt{1 + 4r^2}}\right) \right)$$

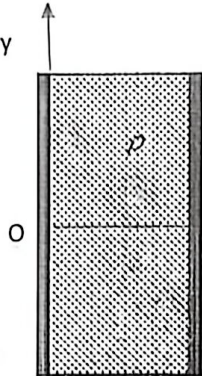
$$\begin{aligned} \text{其中 } \frac{\partial}{\partial r} \ln\left(\frac{-L + \sqrt{L^2 + 4r^2}}{L + \sqrt{L^2 + 4r^2}}\right) &\Rightarrow \frac{\frac{2}{2\sqrt{L^2 + 4r^2}}}{-L + A} \cdot \frac{(-L+A)(L+A)' - (-L+A)'(L+A)}{(L+A)^2} \cdot \frac{8r}{2\sqrt{L^2 + 4r^2}} \\ &= \frac{(-L+A)(\frac{8r}{2\sqrt{L^2 + 4r^2}}) - (\frac{8r}{2\sqrt{L^2 + 4r^2}})(L+A)}{2(A^2 - L^2)\sqrt{L^2 + 4r^2}} \cdot 8r \\ &= \frac{64r^2}{4(L^2 + 4r^2)} \cdot \frac{A^2 - L^2}{2(A^2 - L^2)} = \frac{L}{r \sqrt{\frac{L^2}{4} + r^2}} \end{aligned}$$

$$\therefore \vec{E} = -\nabla(\varphi).$$

2、两块无限大导体平板分别置于 $x=0$ 和 $x=d$ 处，板间充满电荷，其体电荷密度为

$\rho = \frac{\rho_0 x}{d}$ ，极板的电位分别设为 0 和 U_0 ，如图所示，求两导体板之间的电位和电场强度。

解：



由泊松方程 $\nabla \cdot \vec{E} = \nabla \cdot \vec{D} = -\frac{\rho}{\epsilon_0} = -\frac{\rho_0 x}{\epsilon_0 d}$

$$\therefore \frac{d\varphi}{dx} = -\frac{\rho_0}{\epsilon_0 d} \cdot \frac{1}{2} x^2 + C_1$$

$$\therefore \varphi = -\frac{\rho_0}{\epsilon_0 d} \cdot \frac{1}{3} x^3 \cdot \frac{1}{2} + C_1 x + C_2$$

由电位的连续性 $\varphi(x=0) = 0$ ，解得 $C_2 = 0$

$$\varphi(x=d) = -\frac{\rho_0}{\epsilon_0 d} \cdot \frac{1}{6} d^3 + C_1 d = U_0, \text{ 即 } C_1 = \frac{U_0}{d} + \frac{\rho_0}{6\epsilon_0} d$$

$$\therefore \varphi(x) = -\frac{\rho_0 x^3}{6\epsilon_0 d} + \left(\frac{\rho_0 d}{6\epsilon_0} + \frac{U_0}{d} \right) x, \text{ 其中 } 0 \leq x \leq d;$$

$$\vec{E}(\vec{r}) = -\nabla \varphi(x) = -\left[-\frac{3\rho_0 x^2}{6\epsilon_0 d} + \frac{\rho_0 d}{6\epsilon_0} + \frac{U_0}{d} \right] \vec{e}_x, \text{ 其中 } 0 \leq x \leq d;$$

$$= \left[\frac{\rho_0 x^2}{2\epsilon_0 d} - \frac{\rho_0 d}{6\epsilon_0} - \frac{U_0}{d} \right] \vec{e}_x$$

3、平行板电容器的长、宽分别为 a 和 b ，极板间距离为 d ，电容器的另一半厚度 $(0 \leq \frac{d}{2})$ 用

介电常数为 ϵ 的电介质填充，如图所示。

(1) 板上外加电压 U_0 ，求板上的自由电荷面密度、束缚电荷；

(2) 若已知板上的自由电荷总量为 Q ，求此时极板间电压和束缚电荷；

(3) 求电容器的电容量。

解：

$$(2) \quad \rho_s = Q/s = Q/ab$$

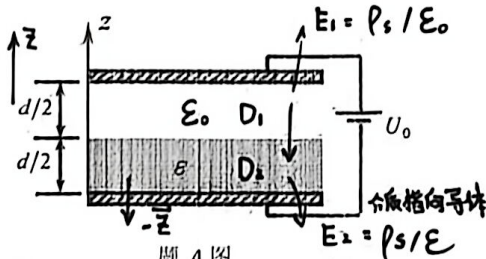
$$U = E_1 d/2 + E_2 d/2 = \frac{\rho_s}{\epsilon_0} \cdot \frac{d}{2} + \frac{\rho_s}{\epsilon} \cdot \frac{d}{2}$$

$$= \frac{\rho_s \cdot d}{2} \left[\frac{\epsilon_0 + \epsilon}{\epsilon_0 \epsilon} \right] = \frac{Q \cdot d}{2ab} \left[\frac{\epsilon_0 + \epsilon}{\epsilon_0 \epsilon} \right]$$

$$\rho_b = -(\epsilon - \epsilon_0) \vec{E}_2 \cdot \vec{e}_z = \frac{\epsilon - \epsilon_0}{\epsilon} \rho_s,$$

$$\rho_b' = \frac{\epsilon - \epsilon_0}{\epsilon} \vec{E}_2 \cdot \vec{e}_z = -\frac{\epsilon - \epsilon_0}{\epsilon} \rho_s,$$

$$(3) \quad C = \frac{Q}{U} = \frac{Q}{\frac{Q \cdot d}{2ab} \left[\frac{\epsilon_0 + \epsilon}{\epsilon_0 \epsilon} \right]} = \frac{2ab \cdot \epsilon_0 \epsilon}{d(\epsilon_0 + \epsilon)};$$



解：由高斯定理 $\nabla \cdot \vec{D} = \rho_s$

$$(1) \quad \therefore \vec{D} \cdot \vec{S} = \rho_s \cdot S \text{ 故 } D = \rho_s$$

$$\text{由 } E_1 = D/\epsilon = D_1/\epsilon_0$$

$$E_2 = D_2/\epsilon, \text{ 得}$$

$$U_0 = \int_0^d \vec{E} \cdot d\vec{l} = \left(\frac{D_1}{\epsilon_0} + \frac{D_2}{\epsilon} \right) \cdot d/2$$

$$\therefore \frac{2U_0}{d} \cdot \frac{\epsilon_0 \epsilon}{\epsilon_0 + \epsilon} = \rho_s$$

$$\therefore \rho_s = \frac{2U_0 \cdot \epsilon_0 \epsilon}{d(\epsilon_0 + \epsilon)},$$

束缚电荷出现在 $z=0$ ， $z=d/2$ ， $z=d$ 的位置上 ($\rho_b = \vec{P} \cdot \vec{n}$)

$$\vec{P}(z=0) = (\epsilon - \epsilon_0) \vec{E} = (\epsilon - \epsilon_0) \cdot \frac{\rho_s}{\epsilon} = \left(1 - \frac{\epsilon_0}{\epsilon} \right) \rho_s;$$

$$\rho_b = +\vec{P}(z=0) \cdot (-\vec{e}_z) = -\left(1 - \frac{\epsilon_0}{\epsilon} \right) \rho_s,$$

$$\vec{P}(z=d/2) = (\epsilon - \epsilon_0) \vec{E} = (\epsilon - \epsilon_0) \cdot \frac{\rho_s}{\epsilon} = \left(1 - \frac{\epsilon_0}{\epsilon} \right) \rho_s;$$

$$\rho_b' = \vec{P}(z=d/2) \cdot (\vec{e}_z) = \left(1 - \frac{\epsilon_0}{\epsilon} \right) \rho_s;$$

4、填充有两层介质的同轴电缆，内导体半径为 a ，外导体内半径为 c ；内外导体之间填充有损耗介质，介质的分界面为同轴圆柱面，分界面半径为 b ，两层介质的介电常数为 ϵ_1 和 ϵ_2 ，电导率为 σ_1 和 σ_2 。设内导体的电压为 U_0 ，外导体接地。求：(1) 介质中的电流密度和电场强度分布；(2) 同轴电缆单位长度的电容及漏电阻。

解：1) $\vec{J} = \sigma \vec{E}$ 本构关系

$$\nabla \cdot \vec{J} = 0 \longrightarrow \oint_S \vec{J} \cdot d\vec{S} = 0,$$



Case 1° $a \leq r < b$ $J \cdot 2\pi r = I$, I 为漏电流, $\vec{J} = \frac{I}{2\pi r} \vec{e}_p$
 Case 2° $b \leq r < c$ 故 $J = I / (2\pi r)$

$$\vec{E}_1 = \vec{J} / \sigma_1 = \frac{I}{2\pi r \sigma_1} \vec{e}_p, \quad a \leq r < b;$$

$$\vec{E}_2 = \vec{J} / \sigma_2 = \frac{I}{2\pi r \sigma_2} \vec{e}_p, \quad b \leq r < c;$$

以下将 I 用 U_0 表示：
$$U_0 = \int_a^b \vec{E}_1 \cdot d\vec{r} + \int_b^c \vec{E}_2 \cdot d\vec{r}$$
$$= \int_a^b \frac{I}{2\pi r \sigma_1} \cdot dr + \int_b^c \frac{I}{2\pi r \sigma_2} \cdot dr$$
$$= \frac{I}{2\pi \sigma_1} \cdot \ln \frac{b}{a} + \frac{I}{2\pi \sigma_2} \cdot \ln \frac{c}{b} \quad (1)$$

$$\therefore I = \frac{U_0}{\frac{1}{2\pi \sigma_1} \ln \frac{b}{a} + \frac{1}{2\pi \sigma_2} \ln \frac{c}{b}} = \frac{2\pi \sigma_1 \sigma_2 U_0}{\sigma_2 \ln \frac{b}{a} + \sigma_1 \ln \frac{c}{b}};$$

$$\therefore \vec{E}_1 = \frac{1}{2\pi r \sigma_1} \cdot \frac{2\pi \sigma_1 \sigma_2 U_0}{\sigma_2 \ln \frac{b}{a} + \sigma_1 \ln \frac{c}{b}} \vec{e}_p = \frac{\sigma_2 U_0}{(\sigma_2 \ln \frac{b}{a} + \sigma_1 \ln \frac{c}{b}) r} \vec{e}_p$$

$$\vec{E}_2 = \frac{1}{2\pi r \sigma_2} \cdot \frac{2\pi \sigma_1 \sigma_2 U_0}{\sigma_2 \ln \frac{b}{a} + \sigma_1 \ln \frac{c}{b}} \vec{e}_p = \frac{\sigma_1 U_0}{(\sigma_2 \ln \frac{b}{a} + \sigma_1 \ln \frac{c}{b}) r} \vec{e}_p$$

$$\vec{J} = \frac{1}{2\pi r} \cdot \frac{2\pi \sigma_1 \sigma_2 U_0}{\sigma_2 \ln \frac{b}{a} + \sigma_1 \ln \frac{c}{b}} \vec{e}_p = \frac{\sigma_1 \sigma_2 U_0}{\sigma_2 \ln \frac{b}{a} + \sigma_1 \ln \frac{c}{b}} \vec{e}_p.$$

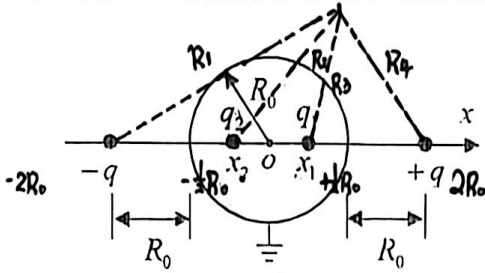
(2) $C = \frac{Q}{U}$ ，令内导体的线电荷密度 ρ_L ，在单位长度上，导体间

$$U_0 \text{ 可表示为 } U_0 = \frac{\rho_L}{2\pi \epsilon_1} \ln \frac{b}{a} + \frac{\rho_L}{2\pi \epsilon_2} \ln \frac{c}{b}$$

$$C = \frac{\rho_L}{U} = \frac{1}{\frac{1}{2\pi \epsilon_1} \ln \frac{b}{a} + \frac{1}{2\pi \epsilon_2} \ln \frac{c}{b}} = \frac{2\pi \epsilon_1 \epsilon_2}{\epsilon_2 \ln \frac{b}{a} + \epsilon_1 \ln \frac{c}{b}}.$$

$$R = \frac{U}{I} = \frac{\sigma_2 \ln \frac{b}{a} + \sigma_1 \ln \frac{c}{b}}{2\pi \sigma_1 \sigma_2}; \quad \text{与 } C \text{ 上形式对称};$$

5、已知接地导体球半径为 R_0 ，在 x 轴上关于原点（球心）对称放置等量异号电荷 $+q$ 、 $-q$ ，位置如图所示。利用镜像法求（1）镜像电荷的位置及电量大小；（2）球外空间电位。



解：（1）推广一般情形，球外电荷距离球心距离为 d ，
镜像电荷距离球心距离为 b

$$\varphi(r) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{R_1} + \frac{q'}{R_2} \right) = 0, \text{ 其中 } \begin{aligned} R_1^2 &= R_0^2 + d^2 - 2xd \cos\theta \\ R_2^2 &= R_0^2 + b^2 - 2xb \cos\theta \end{aligned}$$

代入本题， $\varphi(r) = \frac{1}{4\pi\epsilon_0} \left(\frac{-q}{\sqrt{R_0^2 + 4R_0^2 - 2 \cdot 2R_0 \cdot x_2 \cos\theta}} + \frac{q_2}{\sqrt{R_0^2 + x_2^2 - 2 \cdot R_0 \cdot x_2 \cos\theta}} \right) = 0$

提 $q_2^2 (R_0^2 + x_2^2 - 2R_0 x_2 \cos\theta) = q^2 (5R_0^2 + (-4R_0) \cos\theta)$

$$\begin{cases} 5R_0^2 q_2^2 - (x_2^2 + R_0^2) q^2 = 0 \\ 2R_0 x_2 q_2^2 - 4R_0 q^2 = 0 \end{cases} \Rightarrow \begin{cases} q_2 = \frac{1}{2}q \\ x_2 = -\frac{1}{2}R_0 \end{cases}$$

同理 $\begin{cases} q_1 = -\frac{1}{2}q \\ x_1 = \frac{1}{2}R_0 \end{cases}$ ，综述 $q_1 = -\frac{1}{2}q, q_2 = \frac{1}{2}q$
 $x_1 = \frac{1}{2}R_0, x_2 = -\frac{1}{2}R_0$

（2）球外空间电位由 (x, y) 位置确定，距离分别由 R_1, R_2, R_3, R_4 表示，考虑二维场景下

$$\text{其中 } R_1^2 = y^2 + (x + 2R_0)^2$$

$$R_2^2 = y^2 + (x + \frac{1}{2}R_0)^2$$

$$R_3^2 = y^2 + (x - \frac{1}{2}R_0)^2$$

$$R_4^2 = y^2 + (x - 2R_0)^2$$

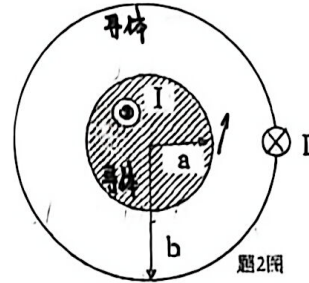
$$\varphi(r) = \frac{1}{4\pi\epsilon_0} \left(\frac{-q}{\sqrt{y^2 + (x + 2R_0)^2}} + \frac{\frac{1}{2}q}{\sqrt{y^2 + (x + \frac{1}{2}R_0)^2}} + \frac{-\frac{1}{2}q}{\sqrt{y^2 + (x - \frac{1}{2}R_0)^2}} + \frac{q}{\sqrt{y^2 + (x - 2R_0)^2}} \right)$$

要考虑三维 (x, y, z) 场景，在 R_1, R_2, R_3, R_4 中引入 z 分量即可。

6、同轴线的内导体半径为 a ，外导体的内半径为 b （外导体的厚度可忽略不计），材料为金属铜，磁导率为 μ_0 ；内、外导体之间填充聚乙烯，其磁导率也为 μ_0 ；设同轴线中的电流为 I ，横截面如图。试求：

(1) 同轴线内导体中以及内、外导体之间的磁场强度分布；

(2) 同轴线单位长度的外自感。



解：(1) case 1° $0 \leq r \leq a$ ，在内导体内部

$$\oint \vec{H} \cdot d\vec{l} = I_1 = J \cdot \pi r^2 = \frac{I}{\pi a^2} \cdot \pi r^2$$

$$\therefore H = \frac{1}{2\pi r} \cdot I \cdot \frac{r^2}{a^2} = I \cdot \frac{r}{2\pi a^2}$$

$$\therefore \vec{H} = I \cdot \frac{r}{2\pi a^2} \cdot \vec{e}_\phi$$

case 2° $a < r \leq b$ ，在内外导体之间

$$\oint \vec{H} \cdot d\vec{l} = I_2 = I - 0 = I$$

$$\therefore H = \frac{I}{2\pi r}, \quad \vec{H} = \frac{I}{2\pi r} \vec{e}_\phi;$$

case 3° $b < r < +\infty$ ，在外导体之外

$$\oint \vec{H} \cdot d\vec{l} = 0, \quad \text{故 } H = 0.$$

(2) 用能量方法 $W_m = \frac{1}{2} L I^2 = \frac{1}{2} \int_V (\vec{B} \cdot \vec{H}) dV$

左边 $W_m = \frac{1}{2} L I^2$ ，其中 I 已知， L 待求

右边 dV 取单位长度圆柱展开 $W_m = \frac{1}{2} \int_a^b (\mu_0 \frac{I}{2\pi r}) \cdot (\frac{I}{2\pi r}) \cdot 1 \cdot 2\pi r dr$

$$= \frac{1}{2} \mu_0 I^2 \int_a^b \frac{1}{2\pi r} dr$$

$$= \frac{\mu_0 I^2}{4\pi} \cdot \ln r \Big|_a^b = \frac{\mu_0 I^2}{4} \ln \frac{b}{a};$$

$$\therefore \frac{1}{2} L I^2 = \frac{1}{4\pi} \mu_0 I^2 \cdot \ln \frac{b}{a}$$

$$\therefore L = \frac{1}{2\pi} \mu_0 \ln \frac{b}{a} = \frac{\mu_0}{2\pi} \ln \frac{b}{a};$$

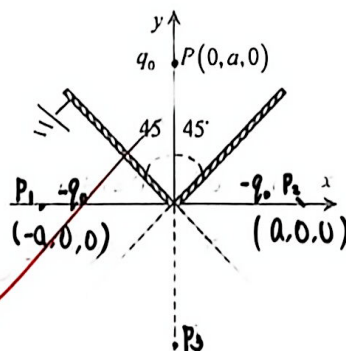
7、如图所示，两块半无限大相互垂直的接地导体平面，在其平分线上 P 点放置一点电荷 q ，求 (1) 各镜像电荷的位置及电量；(2) 两块导体间的电位分布。

解： $\alpha = \frac{\pi}{2}$ ， $n=2$ ， 对称的电荷分量 $2n-1=3$ 个

由对称性 $P_1(-a, 0, 0)$ ， 电量 $-q_0$ 。

$P_2(a, 0, 0)$ ， 电量 $-q_0$ 。

$P_3(0, -a, 0)$ ， 电量 $+q_0$ 。



(2) 导体之间限制了 $y > 0$ ， $y > x$ ， $y > -x$

即 $y > |x|$ 的区域

$$\begin{aligned} \varphi(x, y) = & \frac{q_0}{\sqrt{x^2 + (y-a)^2 + z^2}} \cdot \frac{1}{4\pi\epsilon_0} + \frac{-q_0}{\sqrt{(x+a)^2 + y^2 + z^2}} \cdot \frac{1}{4\pi\epsilon_0} \\ & + \frac{q_0}{\sqrt{x^2 + (y+a)^2 + z^2}} \cdot \frac{1}{4\pi\epsilon_0} + \frac{-q_0}{\sqrt{(x-a)^2 + y^2 + z^2}} \cdot \frac{1}{4\pi\epsilon_0} \end{aligned}$$

其中 $y > |x|$;

第五章 电磁波分析基础

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4.21
d.

1、已知无源的空气中的磁场强度为

$$\vec{H} = \vec{e}_y 0.1 \cos(6\pi \times 10^9 t - kz) \text{ A/m}$$

利用波动方程求常数 k 的值。

解、波动方程 $\nabla^2 \vec{H}(\vec{r}) - \frac{\partial^2 \vec{H}}{\partial t^2} \mu_0 \epsilon_0 = 0$, 其中 $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$

$$\nabla^2 \vec{E}(\vec{r}) - \frac{\partial^2 \vec{E}}{\partial t^2} \mu_0 \epsilon_0 = 0$$

在直角坐标系中, $\nabla^2 \vec{H}(\vec{r}) = \frac{\partial^2 H_x}{\partial x^2} + \frac{\partial^2 H_y}{\partial y^2} + \frac{\partial^2 H_z}{\partial z^2} = -\vec{e}_y 0.1 \cos(\omega t - kz) k^2$, $\omega = 6\pi \times 10^9$

$$\frac{\partial^2 H_y}{\partial t^2} = -\vec{e}_y 0.1 \cos(\omega t - kz) \omega^2$$

于是 $-\vec{e}_y 0.1 \cos(\omega t - kz) k^2 + \vec{e}_y 0.1 \cos(\omega t - kz) \omega^2 \mu_0 \epsilon_0 = 0$

$$-k^2 + (6\pi \times 10^9)^2 \cdot (1.11 \times 10^{-17}) = 0$$

$$\therefore k = 62.8 \text{ rad/m}$$

2、自由空间中的电磁场为

$$\vec{E}(z,t) = \vec{e}_x 100 \cos(\omega t - kz) \quad \text{V/m}$$

$$\vec{H}(z,t) = \vec{e}_y 0.265 \cos(\omega t - kz) \quad \text{A/m}$$

式中 $k = \omega \sqrt{\mu_0 \epsilon_0} = 0.42 \text{ rad/m}$ 。求：

(1) 瞬时坡印廷矢量：

$$\begin{aligned} \text{解：} \vec{S}(z,t) &= \vec{E}(z,t) \times \vec{H}(z,t) \\ &= \vec{e}_x \times 26.5 \times \cos^2(\omega t - kz) \\ &= 26.5 \cos^2(\omega t - kz) \vec{e}_z, \quad k = 0.42 \text{ rad/m}, \quad \omega = \frac{0.42}{\sqrt{\mu_0 \epsilon_0}} = 1.26 \times 10^8 \\ &= 26.5 \cos^2[1.26 \times 10^8 t - 0.42 z] \vec{e}_z; \end{aligned}$$

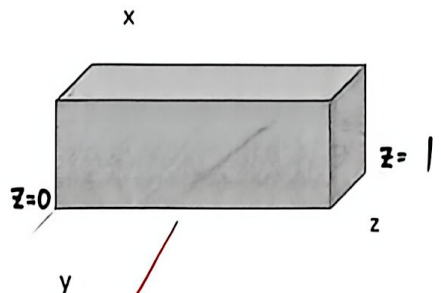
(2) 平均坡印廷矢量：

$$\begin{aligned} \text{解：} S_{av} &= \frac{1}{2} \text{Re} \left(\vec{E}(z) \times \vec{H}^*(z) \right) = \int_0^{\frac{\pi}{2}} 26.5 \cos^2(\omega t - kz) dt \vec{e}_z \\ &= \frac{1}{2} \times 26.5 \vec{e}_z = 13.25 \vec{e}_z \end{aligned}$$

(3) 任一时刻流入如图所示的平行六面体（长1m、横截面积为0.25m²）中的净功率。

$$\text{解：} P = \int_S \vec{S}(z,t) \cdot d\vec{S}$$

$$\begin{aligned} &= + 26.5 \cos^2(\omega t - kz) \times S \Big|_{z=0} \\ &\quad - 26.5 \cos^2(\omega t - kz) \times S \Big|_{z=1} \\ &= 6.625 [\cos^2(\omega t) - \cos^2(\omega t - 0.42)] \\ &= 6.625 [\cos^2(1.26 \times 10^8 t) - \cos^2(1.26 \times 10^8 t - 0.42)]; \end{aligned}$$



4、在横截面为 $a \times b$ 的矩形金属波导中，电磁场的复矢量为

$$\vec{E} = -\vec{e}_y j\omega\mu \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z} \quad \text{V/m} \quad e^{+j\omega t - j\beta z} = \cos(\omega t - \beta z) + j \sin(\omega t - \beta z)$$

$$\vec{H} = [\vec{e}_x j\beta \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) + \vec{e}_z H_0 \cos\left(\frac{\pi x}{a}\right)] e^{-j\beta z} \quad \text{A/m}$$

式中 H_0 、 ω 、 μ 和 β 都是实常数。求：(1) 瞬时坡印廷矢量；

$$\begin{aligned} \vec{E}(r, t) &= \text{Re} \left[-\vec{e}_y j\omega\mu \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) \cdot [\cos(-\beta z) + j \sin(-\beta z)] \cdot [\cos \omega t + j \sin \omega t] \right] \\ &= -\vec{e}_y j\omega\mu \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) \sin(-\beta z) \cos \omega t \\ &\quad - \vec{e}_y j\omega\mu \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) \cos(-\beta z) \sin \omega t \\ &= \vec{e}_y \omega\mu \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) (\sin \omega t \cos \beta z - \cos \omega t \sin \beta z) \\ &= \vec{e}_y \omega\mu \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) \sin(\omega t - \beta z); \\ \vec{H}(r, t) &= \text{Re} \left[\vec{e}_x j\beta \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) + \vec{e}_z H_0 \cos\left(\frac{\pi x}{a}\right) \right] \cdot [\cos(\omega t - \beta z) + j \sin(\omega t - \beta z)] \\ &= \vec{e}_x \beta \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) \sin(\omega t - \beta z) + \vec{e}_z H_0 \cos\left(\frac{\pi x}{a}\right) \cos(\omega t - \beta z) \end{aligned}$$

$$\vec{S}(r, t) = \vec{E}(r, t) \times \vec{H}(r, t) = -\vec{e}_z \omega\mu\beta \frac{a^2}{\pi^2} H_0^2 \sin^2\left(\frac{\pi x}{a}\right) \sin(\omega t - \beta z)$$

(2) 平均坡印廷矢量。

$$\begin{aligned} \vec{S}_a &= \frac{1}{2} \text{Re} [\vec{E}(z) \times \vec{H}^*(z)] = \frac{\text{Re}}{2} \left[\vec{E}(z) \times \left(\vec{e}_x j\beta \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) + \vec{e}_z H_0 \cos\left(\frac{\pi x}{a}\right) \right) \right] \\ &\quad \cos(-\beta z) + j \sin(-\beta z) \\ &= \frac{\text{Re}}{2} \left[-\vec{e}_y j\omega\mu \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z} \right] \times \left[\vec{e}_x j\beta \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) (\cos \beta z - j \sin \beta z) \right. \\ &\quad \left. + \vec{e}_z H_0 \cos\left(\frac{\pi x}{a}\right) (\cos \beta z + j \sin \beta z) \right] \\ &= \frac{\text{Re}}{2} \left[-\vec{e}_y j\omega\mu \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) e^{-j\beta z} \times \left(-\vec{e}_x j\beta \frac{a}{\pi} H_0 \sin\left(\frac{\pi x}{a}\right) + \vec{e}_z H_0 \cos\left(\frac{\pi x}{a}\right) \right) e^{j\beta z} \right] \\ &= \frac{1}{2} \cdot j^2 \cdot (-\vec{e}_z) \omega\mu\beta \cdot \left(\frac{a}{\pi} H_0\right)^2 \cdot \frac{1}{2} \sin^2\left(\frac{\pi x}{a}\right) \\ &= \frac{1}{2} \vec{e}_z \cdot \omega\mu\beta \cdot \left(\frac{a}{\pi} H_0\right)^2 \sin^2\left(\frac{\pi x}{a}\right); \end{aligned}$$

3、已知某电磁场的复矢量为

$$\vec{E}(z) = \vec{e}_x j E_0 \sin(k_0 z) \quad \text{V/m}$$

$$\vec{H}(z) = \vec{e}_y \sqrt{\frac{\epsilon_0}{\mu_0}} E_0 \cos(k_0 z) \quad \text{A/m}$$

式中 $k_0 = \frac{2\pi}{\lambda_0} = \frac{\omega}{c}$, c 为真空中的光速, λ_0 是波长。求: $k_0 \lambda_0 = 2\pi$

(1) $z=0$ 、 $\frac{\lambda_0}{8}$ 、 $\frac{\lambda_0}{4}$ 各点处的瞬时坡印廷矢量:

$$\begin{aligned} \vec{E}(z, t) &= \operatorname{Re} [\vec{e}_x j E_0 \sin(k_0 z) e^{j\omega t}] = \operatorname{Re} [\vec{e}_x j E_0 \sin(k_0 z) (\cos \omega t + j \sin \omega t)] \\ &= -\vec{e}_x E_0 \sin(k_0 z) \sin \omega t \end{aligned}$$

$$\begin{aligned} \vec{H}(z, t) &= \operatorname{Re} [\vec{e}_y \sqrt{\frac{\epsilon_0}{\mu_0}} E_0 \cos(k_0 z) e^{j\omega t}] = \operatorname{Re} [\vec{e}_y \sqrt{\frac{\epsilon_0}{\mu_0}} E_0 \cos(k_0 z) (\cos \omega t + j \sin \omega t)] \\ &= \vec{e}_y E_0 \cos(k_0 z) \cos \omega t \sqrt{\frac{\epsilon_0}{\mu_0}} \end{aligned}$$

$$\therefore \vec{S}(z, t) = \vec{E}(z, t) \times \vec{H}(z, t) = -\vec{e}_z \cdot \frac{1}{4} \cdot \sin(2k_0 z) \sin(2\omega t) \cdot \sqrt{\frac{\epsilon_0}{\mu_0}}$$

$$\vec{S}(0, t) = -\vec{e}_z \cdot \frac{1}{4} \cdot 0 \cdot \sin(2\omega t) = 0;$$

$$\vec{S}\left(\frac{\lambda_0}{8}, t\right) = -\vec{e}_z \cdot \frac{1}{4} \cdot \sin\left(\frac{k_0 \lambda_0}{4}\right) \sin(2\omega t) \sqrt{\frac{\epsilon_0}{\mu_0}} = -\vec{e}_z \cdot \frac{1}{4} \cdot \sin(2\omega t) \cdot \sqrt{\frac{\epsilon_0}{\mu_0}}$$

$$\vec{S}\left(\frac{\lambda_0}{4}, t\right) = -\vec{e}_z \cdot \frac{1}{4} \cdot \sin\left(\frac{k_0 \lambda_0}{2}\right) \sin(2\omega t) \sqrt{\frac{\epsilon_0}{\mu_0}} = 0.$$

(2) 以上各点处的平均坡印廷矢量。

$$\begin{aligned} \vec{S}_{av} &= \frac{1}{2} \operatorname{Re} [\vec{E}(z) \times \vec{H}^*(z)] \\ &= \frac{1}{2} \operatorname{Re} [\vec{e}_x j E_0 \sin(k_0 z) \times \vec{e}_y \sqrt{\frac{\epsilon_0}{\mu_0}} E_0 \cos(k_0 z)] \\ &= \frac{1}{2} \vec{e}_z \cdot \operatorname{Re} [j E_0^2 \sin(k_0 z) \cdot \sqrt{\frac{\epsilon_0}{\mu_0}} \cos(k_0 z)] \\ &= 0; \end{aligned}$$

5、已知真空中时变场的矢量磁位为 $\vec{A}(z,t) = \vec{e}_x A_0 \cos(\omega t - kz)$ ，试求：

(1) 电场强度 $\vec{E}(z,t)$ 和磁场强度 $\vec{H}(z,t)$ ；

(2) 坡印廷矢量及其平均值。

$$\text{解：} \vec{B} = \nabla \times \vec{A} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_0 \cos(\omega t - kz) & 0 & 0 \end{vmatrix} = \begin{vmatrix} \vec{e}_x & \vec{e}_y \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ A_0 \cos(\omega t - kz) & 0 \end{vmatrix}$$

$$\vec{B}(z,t) = \vec{e}_y A_0 \sin(\omega t - kz) \cdot (-k) = \vec{e}_y A_0 k \sin(\omega t - kz)$$

$$\vec{H}(z,t) = \vec{B}(z,t) / \mu_0 = \frac{A_0 k}{\mu_0} \sin(\omega t - kz) \vec{e}_y ;$$

$$\text{由 } \nabla \times \vec{E}(z,t) = -\frac{\partial \vec{B}}{\partial t} = -\vec{e}_y A_0 k \cos(\omega t - kz) \cdot \omega ;$$

$$\nabla \times \vec{H}(z,t) = \frac{\partial \vec{D}}{\partial t} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \# \nabla \times \vec{H}(z,t) = \vec{e}_x \frac{A_0 k^2}{\mu_0} \cos(\omega t - kz)$$

$$\therefore \vec{E}(z,t) = \vec{e}_x \cdot \frac{A_0 k^2}{\mu_0 \epsilon_0} \cdot \frac{1}{\omega} \sin(\omega t - kz)$$

$$\text{注意到 } k = \omega \sqrt{\mu_0 \epsilon_0}, \text{ 即 } k^2 / (\mu_0 \epsilon_0) = \omega^2.$$

$$\therefore \vec{E}(z,t) = \vec{e}_x A_0 \omega \sin(\omega t - kz);$$

$$\begin{aligned} (2) \vec{E}(z,t) \times \vec{H}(z,t) &= \frac{A_0 k}{\mu_0} \sin(\omega t - kz) \cdot \omega \sin(\omega t - kz) \vec{e}_z \\ &= \vec{e}_z \cdot A_0^2 \cdot \frac{k\omega}{\mu_0} \sin^2(\omega t - kz); \end{aligned}$$

$$\vec{S}_{av} = \frac{1}{2} \text{Re} [\vec{E}(z,t) \times \vec{H}^*(z,t)] = \frac{1}{2} \text{Re} [\vec{E}(z,t) \times \vec{H}(z,t)]$$

$$= \frac{1}{2} \text{Re} \left[\vec{e}_z \cdot A_0^2 \cdot \frac{k\omega}{\mu_0} \sin^2(\omega t - kz) \right]$$

$$\vec{S}_{av} = \int_0^T \vec{S}(z,t) dt = \vec{e}_z \cdot A_0^2 \cdot \frac{k\omega}{\mu_0} \cdot \frac{1}{2};$$

第六章 均匀平面波的空间传播分析

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1、在自由空间中已知电场 $\vec{E}(z, t) = \vec{e}_y 10^3 \sin(\omega t - \beta z)$ V/m, 试求出磁场强度 $\vec{H}(z, t)$ 。

$$\begin{aligned} \text{解: } \vec{H}(z, t) &= \frac{1}{Z_0} \vec{e}_z \times \vec{E}(z, t) = \frac{1}{377} (\vec{e}_z \times \vec{e}_y 10^3 \sin(\omega t - \beta z)) \\ &= \frac{1}{377} [-\vec{e}_x 10^3 \sin(\omega t - \beta z)] \text{ A/m} \\ &= -2.65 \vec{e}_x \sin(\omega t - \beta z) \text{ A/m} \end{aligned}$$

2、若真空中正弦电磁场的电场复矢量为

$$\vec{E}(\vec{r}) = (-j\vec{e}_x - 2\vec{e}_y + j\sqrt{3}\vec{e}_z) e^{-j0.05\pi(\sqrt{3}x+z)} \text{ V/m}$$

试求磁感应强度的复矢量 $\vec{B}(\vec{r})$ 及平均能流密度矢量 $\vec{S}_{av}$ 。

$$\begin{aligned} \text{解: } \vec{H}(\vec{r}) &= \frac{1}{Z_0} [\vec{e}_z \times \vec{E}(\vec{r})] = \frac{1}{377} [-j\vec{e}_y + 2\vec{e}_x] e^{-j0.05\pi(\sqrt{3}x+z)} \\ \vec{B}(\vec{r}) &= \mu_0 \vec{H}(\vec{r}) = \frac{\mu_0}{377} [-j\vec{e}_y + 2\vec{e}_x] e^{-j0.05\pi(\sqrt{3}x+z)} \\ \therefore \vec{S}_{av} &= \frac{1}{2} [\vec{E}(\vec{r}) \times \vec{H}^*(\vec{r})] \\ &= \frac{\mu_0}{2} [(-j\vec{e}_x - 2\vec{e}_y + j\sqrt{3}\vec{e}_z) e^{-j0.05\pi(\sqrt{3}x+z)} \\ &\quad \times (\frac{1}{377} (j\vec{e}_y + 2\vec{e}_x) e^{j0.05\pi(\sqrt{3}x+z)})] \\ &= \frac{\mu_0}{2 \times 377} [-\vec{e}_z - \sqrt{3}\vec{e}_x + 4\vec{e}_z] \\ &= \frac{1}{754} [-\sqrt{3}\vec{e}_x + 3\vec{e}_z] \end{aligned}$$

3、在自由空间传播的均匀平面波的电场强度复矢量为

$$\vec{E} = \vec{e}_x 10^{-4} e^{-j20\pi z} + \vec{e}_y 10^{-4} e^{-j(20\pi z - \frac{\pi}{2})} \text{ V/m}$$

试求：

- (1) 平面波的传播方向和频率。
- (2) 波的极化方式；
- (3) 磁场强度 $\vec{H}(z, t)$ ；
- (4) 流过与传播方向垂直的单位面积的平均功率。

解：(1) $\vec{E} = (\vec{e}_x + j\vec{e}_y) \times 10^{-4} \times e^{-j20\pi z}$

传播方向 z 轴 ($\vec{e}_z$)，方向

由 $k\lambda = 2\pi$ ，得 $\lambda = 0.1\text{m}$ ，故 $f = \frac{c}{\lambda} = 3 \times 10^9 \text{ Hz}$

(2) 由 $|\vec{E}_x| = |\vec{E}_y|$ ，相位差 $\frac{\pi}{2}$ ，圆极化

$$\vec{E}(\vec{r}, t) = \text{Re}[(\vec{e}_x + j\vec{e}_y) \times 10^{-4} \times e^{j(\omega t - 20\pi z)}]$$

$$= [\vec{e}_x \cos(\omega t - 20\pi z) + \vec{e}_y \cos(\omega t + \frac{\pi}{2} - 20\pi z)] \times 10^{-4}$$

当 $\omega t = 0$ 时， $E_x = \cos 0 = 1$ ； $E_y = \cos \frac{\pi}{2} = 0$ ；

当 $\omega t = \frac{\pi}{2}$ 时， $E_x = \cos \frac{\pi}{2} = 0$ ； $E_y = \cos \pi = -1$ ；

左旋圆极化

(3) $\vec{H}(\vec{r}, t) = \frac{1}{Z} (\vec{e}_z \times \vec{E}(\vec{r}, t))$

$$= \frac{1}{377} (\vec{e}_y \cos(\omega t - 20\pi z) + (-\vec{e}_x) \cos(\omega t + \frac{\pi}{2} - 20\pi z)) \times 10^{-4}$$

$$= \frac{1}{377} (\vec{e}_y \cos(\omega t - 20\pi z) - \vec{e}_x \cos(\omega t + \frac{\pi}{2} - 20\pi z)) \times 10^{-4}$$

(4) $S_{av} = \frac{1}{2} [\vec{E}(\vec{r}) \times \vec{H}^*(\vec{r})] = \frac{\text{Re}}{2} [(\vec{e}_x + j\vec{e}_y) \times 10^{-4} \times e^{-j20\pi z}]$

$$\times [(\vec{e}_y + j\vec{e}_x) \times 10^{-4} \times e^{j20\pi z}] \times \frac{1}{377}$$

$$= \frac{\text{Re}}{2} [(\vec{e}_z + \vec{e}_z) \times 10^{-8}] \times \frac{1}{377} = 2.65 \times 10^{-11} \vec{e}_z \text{ W/m}^2$$

平均功率 $P = S_{av} \cdot S = 2.65 \times 10^{-11} \text{ W/m}^2 \times 1 \text{ m}^2 = 2.65 \times 10^{-11} \text{ W}$;

4、已知自由空间传播的均匀平面波的磁场强度为

$$\vec{H} = \left(\vec{e}_x \frac{3}{2} + \vec{e}_y + \vec{e}_z \right) 10^{-6} \cos \left[\omega t - \pi \left(-x + y + \frac{1}{2}z \right) \right] \text{ A/m}$$

试求：

$$\vec{H}(\vec{r}, t) = \cos [\omega t + \pi (x - y - \frac{1}{2}z)] = \cos [\omega t + \vec{k} \cdot \vec{r}]$$

- (1) 波的传播方向；
- (2) 波的频率和波长；
- (3) 与磁场相伴的电场 $\vec{E}(\vec{r})$ ；
- (4) 平均坡印廷矢量。

解：(1) $\vec{H}(\vec{r}, t) = \text{Re} [\vec{H}(\vec{r}) \cdot e^{j\omega t}]$, 故 $\vec{H}(\vec{r}) = (\vec{e}_x \frac{3}{2} + \vec{e}_y + \vec{e}_z) \times 10^{-6} \times e^{j\pi(x-y-\frac{1}{2}z)}$

波传播方向 $\vec{e}_\omega = [-\vec{e}_x + \vec{e}_y + \frac{1}{2}\vec{e}_z] / \sqrt{1^2 + 1^2 + \frac{1}{4}} = -\frac{2}{3}\vec{e}_x + \frac{2}{3}\vec{e}_y + \frac{1}{3}\vec{e}_z$

(2) $\vec{k} \cdot \vec{r} = [\pi \vec{e}_x + (-\pi) \vec{e}_y + (-\frac{\pi}{2}) \vec{e}_z] \cdot (x \vec{e}_x + y \vec{e}_y + z \vec{e}_z)$

于是 $k = |\vec{k}| = \pi \sqrt{1^2 + 1^2 + \frac{1}{4}} = \frac{3}{2}\pi$

$\therefore k\lambda = 2\pi$ 故 $\lambda = \frac{2\pi}{k} = \frac{2}{3/2} = \frac{4}{3} \text{ m}$

$\therefore c = \lambda f = 3 \times 10^8$ 故 $f = \frac{3 \times 10^8}{\frac{4}{3}} \text{ Hz} = \frac{9}{4} \times 10^8 \text{ Hz}$

(3) $\vec{E}(\vec{r})$ 由 $\vec{H}(\vec{r}) = \frac{1}{Z} \vec{e}_\omega \times \vec{E}(\vec{r})$ 可知: $\vec{E}(\vec{r}) = Z \cdot \vec{H}(\vec{r}) \times \vec{e}_\omega$

$$\begin{aligned} \vec{E}(\vec{r}) &= \frac{120\pi}{1} (\vec{e}_x \cdot \frac{3}{2} + \vec{e}_y + \vec{e}_z) \times 10^{-6} \times e^{j\pi(x-y-\frac{1}{2}z)} \times \vec{e}_\omega \\ &= [1.2\pi \times 10^{-4}] (\vec{e}_x \cdot \frac{3}{2} + \vec{e}_y + \vec{e}_z) \times (-\frac{2}{3}\vec{e}_x + \frac{2}{3}\vec{e}_y + \frac{1}{3}\vec{e}_z) \times e^{j\pi(x-y-\frac{1}{2}z)} \\ &= [1.2\pi \times 10^{-4}] (\vec{e}_x \cdot (-\frac{1}{3}) + \vec{e}_y \cdot (-\frac{7}{6}) + \vec{e}_z \cdot (\frac{5}{3})) \times e^{j\pi(x-y-\frac{1}{2}z)} \end{aligned}$$

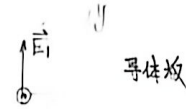
(4) $S_{av} = \frac{\text{Re}}{2} [\vec{E}(\vec{r}) \times \vec{H}^*(\vec{r})]$

$$\begin{aligned} &= \frac{\text{Re}}{2} [1.2\pi \times 10^{-4} (\vec{e}_x \cdot (-\frac{1}{3}) + \vec{e}_y \cdot (-\frac{7}{6}) + \vec{e}_z \cdot (\frac{5}{3})) \\ &\quad \times 1 \times 10^{-6} (\vec{e}_x \cdot \frac{3}{2} + \vec{e}_y + \vec{e}_z)] \\ &= 0.6\pi \times 10^{-10} \cdot \text{Re} [-\frac{1}{6}\vec{e}_x + \frac{1}{6}\vec{e}_y + \frac{1}{12}\vec{e}_z] \\ &= 0.6\pi \times 10^{-10} (-\frac{1}{6}\vec{e}_x + \frac{1}{6}\vec{e}_y + \frac{1}{12}\vec{e}_z) \end{aligned}$$

5、有一频率为 100MHz 、沿 y 方向极化的均匀平面波从空气 ($x < 0$ 区域) 中垂直入射到位于 $x = 0$ 的理想导体板上。设入射波电场 $\vec{E}_i$ 的振幅为 10V/m ，试求：

(1) 合成波电场 $\vec{E}_1$ 和磁场 $\vec{H}_1$ 的复矢量；

解：(1) $\vec{E}_i = 10\vec{e}_y \exp(-jkx)$ ，其中



$$k\lambda = 2\pi; \quad c = \lambda f, \quad \text{即 } k = \frac{2\pi}{\lambda} = \frac{2\pi}{c/f} = \frac{2\pi}{3 \times 10^8} \times 100 \times 10^6 = \frac{2}{3}\pi$$

$$\therefore \vec{E}_i = 10\vec{e}_y \exp(-j\frac{2}{3}\pi x), \quad \vec{E}_r = 10\vec{e}_y \Gamma \exp(+j\frac{2}{3}\pi x), \quad \text{其中 } \Gamma = -1$$

$$\begin{aligned} \vec{H}_i &= \frac{1}{\epsilon_0} \vec{e}_\omega \times \vec{E}_i = \frac{1}{120\pi} \vec{e}_x \times (10\vec{e}_y \exp(-j\frac{2}{3}\pi x)) \\ &= \frac{1}{12\pi} \vec{e}_z \cdot \exp(-j\frac{2}{3}\pi x); \quad \vec{H}_r = \frac{1}{12\pi} \vec{e}_z \Gamma \exp(+j\frac{2}{3}\pi x); \end{aligned}$$

$$\therefore \vec{E}_1 = \vec{E}_i + \vec{E}_r = 10\vec{e}_y (\exp(-j\frac{2}{3}\pi x) - \exp(+j\frac{2}{3}\pi x)) = -20j\vec{e}_y (\sin \frac{2}{3}\pi x);$$

$$\begin{aligned} \vec{H}_1 &= \vec{H}_i + \vec{H}_r = \frac{1}{12\pi} \vec{e}_z \cdot \exp(-j\frac{2}{3}\pi x) - \frac{1}{12\pi} \vec{e}_z \cdot \Gamma \cdot \exp(+j\frac{2}{3}\pi x) \\ &= \frac{1}{12\pi} \vec{e}_z \cdot (\exp(-j\frac{2}{3}\pi x) + \exp(+j\frac{2}{3}\pi x)); \\ &= \frac{1}{12\pi} \vec{e}_z \cdot 2 \cos(\frac{2}{3}\pi x) = \frac{1}{6\pi} \vec{e}_z \cos(\frac{2}{3}\pi x) \end{aligned}$$

(2) 距离导体平面最近的合成波电场 $\vec{E}_1$ 为 0 的位置。

$$(2) \vec{E}_1 = 10\vec{e}_y (\exp(-j\frac{2}{3}\pi x) - \exp(+j\frac{2}{3}\pi x)) = 0$$

$$\therefore 10\vec{e}_y \cdot 2j \cdot \sin \frac{2}{3}\pi x = 0$$

$$\therefore \frac{2}{3}\pi x = k\pi, \quad k=0, \pm 1, \pm 2, \dots$$

$$\therefore x = \frac{3}{2}k, \quad k=0, \pm 1, \pm 2, \dots \text{ 且在 } x < 0 \text{ 区域内}$$

$$\therefore \text{最近位置为 } k=-1, \quad x = -\frac{3}{2} \text{ 处。}$$

6、一均匀平面波沿+z方向传播，其电场强度矢量为

$$\vec{E} = \vec{e}_x 120 \sin(\omega t - \beta z) + \vec{e}_y 240 \cos(\omega t - \beta z) \text{ V/m}$$

(1) 应用麦克斯韦方程求相伴的磁场 $\vec{H}$;

$$\begin{aligned} \text{解: } \vec{E}(\vec{r}, t) &= \vec{e}_x 120 \sin(\omega t - \beta z) + \vec{e}_y 240 \cos(\omega t - \beta z) = \operatorname{Re} \left\{ \vec{e}_x 120 e^{j(\omega t - \beta z + \frac{\pi}{2})} + \vec{e}_y 240 e^{j(\omega t - \beta z)} \right\} \\ \vec{E}(\vec{r}, t) &= \operatorname{Re} \left\{ \vec{E}(\vec{r}) e^{j\omega t} \right\} \Rightarrow \vec{e}_x 120 e^{j(-\beta z + \frac{\pi}{2})} + \vec{e}_y 240 e^{j(-\beta z)} = \vec{E}(\vec{r}) \\ \therefore \vec{H}(\vec{r}) &= \frac{1}{\eta} \vec{e}_z \times \vec{E}(\vec{r}) = \frac{1}{\eta} \vec{e}_z \times \left[\vec{e}_x 120 e^{j(-\beta z + \frac{\pi}{2})} + \vec{e}_y 240 e^{j(-\beta z)} \right] \\ &= \frac{1}{120\pi} \left[\vec{e}_y 120 e^{j(-\beta z + \frac{\pi}{2})} + -\vec{e}_x 240 e^{j(-\beta z)} \right] \\ &= \frac{1}{\pi} \left[\vec{e}_y e^{j(-\beta z + \frac{\pi}{2})} - \vec{e}_x 2 e^{j(-\beta z)} \right] \text{ A/m} \end{aligned}$$

(2) 若在波传播方向上 $z = 0$ 处，放置一无限大的理想导体板，求 $z < 0$ 区域中的 $\vec{E}_1$ 和 $\vec{H}_1$ ；(合成波方程式)

$$\begin{aligned} \text{解: } \vec{E}_1 &= \vec{E}_i + \vec{E}_r, \text{ 其中 } \vec{E}_i = \vec{e}_x 120 e^{j(-\beta z + \frac{\pi}{2})} + \vec{e}_y 240 e^{j(-\beta z)} \\ \vec{E}_r &= r \left[\vec{e}_x 120 e^{j(\beta z - \frac{\pi}{2})} + \vec{e}_y 240 e^{j(\beta z)} \right], r = -1 \\ \therefore \vec{E}_1 &= \vec{e}_x 120 [e^{j(-\beta z + \frac{\pi}{2})} - e^{j(\beta z - \frac{\pi}{2})}] + \vec{e}_y 240 [e^{j(-\beta z)} - e^{j(\beta z)}]; \\ \vec{H}_1 &= \vec{H}_i + \vec{H}_r, \text{ 其中 } \vec{H}_i = \vec{H}(\vec{r}) = \frac{1}{\pi} [\vec{e}_y e^{j(-\beta z + \frac{\pi}{2})} - \vec{e}_x 2 e^{j(-\beta z)}] \\ \vec{H}_r &= \frac{1}{\pi} r [\vec{e}_y e^{j(\beta z - \frac{\pi}{2})} - \vec{e}_x 2 e^{j(\beta z)}], r = -1 \\ \therefore \vec{H}_1 &= \frac{1}{\pi} [\vec{e}_y (e^{j(-\beta z + \frac{\pi}{2})} + e^{j(\beta z - \frac{\pi}{2})}) - \vec{e}_x (2 e^{-j\beta z} + 2 e^{j\beta z})] \end{aligned}$$

$$\begin{aligned} \text{综述, } \vec{E}_1 &= -\vec{e}_x 120 \cdot 2j \sin(\beta z - \frac{\pi}{2}) - \vec{e}_y 240 \cdot 2j \sin(\beta z) \\ &= -240j \vec{e}_x \cos \beta z - 480j \vec{e}_y \sin \beta z \text{ V/m} \end{aligned}$$

$$\vec{H}_1 = \frac{1}{\pi} [\vec{e}_y \cos(\beta z - \frac{\pi}{2}) - 2\vec{e}_x \cos \beta z] \text{ A/m};$$

(3) 求理想导体板表面的电流密度。

$$\begin{aligned} \text{解: } \vec{J} &= \vec{e}_z \times \vec{H}_1(z=0) \\ &= \vec{e}_z \times [\vec{e}_y \cdot 0 - 2\vec{e}_x] = -2\vec{e}_y \end{aligned}$$

7、一圆极化波自空气中垂直入射于一理想介质板上，介质板的本征阻抗为 η_2 。入射波电场为 $\vec{E} = E_m(\vec{e}_x + \vec{e}_y j)e^{-j\beta z}$ ，求反射波与透射波的电场，它们的极化情况如何？

解：本征阻抗 η_1 已改为 Z_1 ， $Z_1 = Z_0 = 120\pi$

$$\vec{E}_i = E_m(\vec{e}_x + \vec{e}_y j)e^{-j\beta z}, \text{ 其中 } \beta = \omega\sqrt{\mu_0\epsilon_0}$$

$$\vec{E}_r = E_m\Gamma(\vec{e}_x + \vec{e}_y j)e^{+j\beta z}, \text{ 其中 } \Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{\eta_2 - \eta_0}{\eta_2 + \eta_0}$$

$$\vec{E}_t = E_m\tau(\vec{e}_x + \vec{e}_y j)e^{-jkz}, \text{ 其中 } \tau = 1 + \frac{\eta_2 - \eta_0}{\eta_2 + \eta_0} = \frac{2\eta_2}{\eta_2 + \eta_0}$$

$$k = \omega\sqrt{\mu_0\epsilon_r}\sqrt{\mu_0\epsilon_0}$$

$$\text{反射波 } \vec{E}_r = E_m \cdot \frac{\eta_2 - \eta_0}{\eta_2 + \eta_0} [\vec{e}_x \cdot e^{j\beta z} + \vec{e}_y \cdot e^{j(\beta z + \frac{\pi}{2})}], \text{ 圆极化}$$

$$\vec{E}_r(\vec{r}, t) = \text{Re}[\vec{E}_r(\vec{r})e^{j\omega t}] = E_m \cdot \frac{\eta_2 - \eta_0}{\eta_2 + \eta_0} [\vec{e}_x \cos(\omega t + \beta z) + \vec{e}_y \cos(\omega t + \beta z + \frac{\pi}{2})]$$

$$\text{当 } \omega t = 0 \text{ 时 } E_x = E_m \cdot \frac{\eta_2 - \eta_0}{\eta_2 + \eta_0} \cdot \cos 0, E_y = 0$$

$$\text{当 } \omega t = \frac{\pi}{2} \text{ 时 } E_x = E_m \cdot \frac{\eta_2 - \eta_0}{\eta_2 + \eta_0} \cdot \cos \frac{\pi}{2} = 0, E_y = -E_m \cdot \frac{\eta_2 - \eta_0}{\eta_2 + \eta_0}, \text{ 右旋圆极化}$$

$$\text{折射波 } \vec{E}_t = E_m \cdot \frac{2\eta_2}{\eta_2 + \eta_0} [\vec{e}_x e^{-jkz} + \vec{e}_y e^{j(\frac{\pi}{2} - kz)}], \text{ 圆极化}$$

$$\vec{E}_t(\vec{r}, t) = \text{Re}[\vec{E}_t(\vec{r})e^{j\omega t}] = E_m \cdot \frac{2\eta_2}{\eta_2 + \eta_0} [\vec{e}_x \cos(\omega t - kz) + \vec{e}_y \sin(\omega t - kz)]$$

8、均匀平面波从空气垂直入射到某理想介质平面时，空气中的驻波比为2.7，介质平面上为驻波电场最小点，求电介质的相对介电常数。

$$\text{(解)} \quad z=0 \quad \text{当 } \omega t=0 \text{ 时 } E_x = E_m \cdot \frac{2\eta_2}{\eta_2 + \eta_0}$$

$$\text{当 } \omega t = \frac{\pi}{2} \text{ 时 } E_x = 0$$

$$E_y = 0$$

$$E_y = E_m \frac{2\eta_2}{\eta_2 + \eta_0}$$

左旋圆极化

$$\text{8. 解: } S = \frac{1 + |\Gamma|}{1 - |\Gamma|} = 2.7, \text{ 故 } |\Gamma| = 0.46$$

由介质上驻波电场最小点， $\Gamma = -0.46$

$$\text{由 } \Gamma = \frac{Z_2 - Z_1}{Z_2 + Z_1} = -0.46, Z_2 = \sqrt{\frac{\mu_0}{\epsilon_r}} \cdot 120\pi = 139.43, \text{ 即 } \frac{\mu_r}{\epsilon_r} = 0.136$$

$$\text{由 } \mu_r = 1, \text{ 知 } \epsilon_r = 7.35;$$

7.27.

9、有一厚度为 d ，本质阻抗为 η_2 的介质置于本质阻抗分别为 η_1 和 η_3 的媒质之间，并且 $d = \frac{1}{4}\lambda_2$ ，欲使均匀平面波从媒质 1 垂直入射于媒质 2 的分界面上时不发生反射。求： η_2 、 η_1 和 η_3 应满足什么条件，写出完整的推导过程。

解、将媒质 2，媒质 3 作 Z_{ef} 等效，最简单的一个式子求解

$$Z_{ef} = \eta_2 \frac{\eta_3 + j\eta_2 \tan(\beta_2 d)}{\eta_2 + j\eta_3 \tan(\beta_2 d)}, \quad \beta_2 d = \frac{2\pi}{\lambda_2} \times d = \frac{2\pi}{\lambda_2} \times \frac{1}{4}\lambda_2 = \frac{\pi}{2}$$

$$= \eta_2 \frac{\eta_3 + j\eta_2 \cdot \infty}{\eta_2 + j\eta_3 \cdot \infty} \xrightarrow{\text{除以 } j\eta_3} \frac{\eta_2^2}{\eta_3}$$

由无反射，即 $\Gamma = \frac{Z_{ef} - \eta_1}{Z_{ef} + \eta_1} = 0$ ，即 $Z_{ef} = \frac{\eta_2^2}{\eta_3} = \eta_1$

$$\therefore \eta_2 = \sqrt{\eta_1 \eta_3}$$

* $\vec{E}_i(x, z) = \vec{e}_y 10 e^{-j(6x+8z)}$ 2025-2026 学年第二学期《电磁场与电磁波》作业

10、有一正弦均匀平面波由空气斜入射到位于 $z=0$ 的理想导体平面上，其电场强度的复数形式为 $\vec{E}_i(x, z) = \vec{e}_y 10 e^{-j(6x+8z)}$ V/m。试求：

(1) 入射波的频率 f 与波长 λ ；

解： $\vec{k} \cdot \vec{r} = 6x + 8z$, $\vec{k} = 6\vec{e}_x + 8\vec{e}_z$, $\vec{e}_n = \frac{\vec{k}}{k} = 0.6\vec{e}_x + 0.8\vec{e}_z$

$\therefore k\lambda = 2\pi$ 故波长 $\lambda = \frac{2\pi}{k} = \frac{\pi}{5}$ m;

$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{\pi/5} = 4.77 \times 10^8$ Hz;

(2) $\vec{E}_i(x, z, t)$ 和 $\vec{H}_i(x, z, t)$ 的瞬时表达式；

解： $\vec{E}_i(x, z, t) = \text{Re}[\vec{E}_i(x, z) e^{j\omega t}] = \vec{e}_y 10 \cos(\omega t - 6x - 8z)$ V/m

$\vec{H}_i(x, z, t) = \frac{1}{Z_0} \vec{e}_n \times [\vec{e}_y 10 \cos(\omega t - 6x - 8z)]$
 $= \frac{1}{120\pi} (-\vec{e}_x 0.8 + \vec{e}_z 0.6) 10 \cos(\omega t - 6x - 8z)$
 $= \frac{1}{120\pi} (-\vec{e}_x 8 + \vec{e}_z 6) \cos(\omega t - 6x - 8z)$ A/m

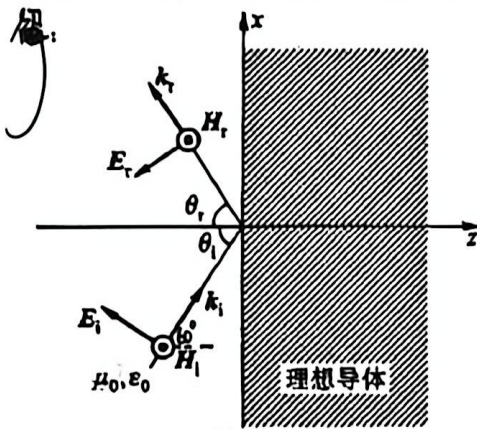
(3) 入射角 θ_i ；

解：图解法
 $\tan \theta_i = \frac{6}{8} = \frac{3}{4}$
 $\therefore \theta_i = \arctan \frac{3}{4} = 37^\circ$

(4) 反射波的 $\vec{E}_r(x, z)$ 和 $\vec{H}_r(x, z)$ 。 $\Gamma = -1$; $\vec{k} = 6\vec{e}_x + 8\vec{e}_z$, 在 $z=0$ 处

解： $\vec{E}_r(x, z) = -\vec{e}_y 10 e^{-j(6x+8z)} = -\vec{e}_y 10 e^{-j(6x+8z)}$
 $\vec{H}_r(x, z) = \frac{1}{Z_0} \vec{e}_n \times \vec{E}_r(x, z) = \frac{1}{120\pi} (6\vec{e}_x - 8\vec{e}_z) \times \vec{E}_r(x, z)$
 $= \frac{1}{120\pi} (-8\vec{e}_x - 6\vec{e}_z) e^{-j(6x+8z)}$

11、频率 $f = 100\text{MHz}$ 的平行极化正弦均匀平面波，在空气 ($z < 0$ 的区域) 中以入射角 $\theta_i = 60^\circ$ 斜入射到 $z = 0$ 处的理想导体表面。设入射波磁场的振幅为 0.1A/m 、方向为 y 方向，如图所示。(1) 求出入射波、反射波的电场和磁场表达式；



$$(1) \vec{E}_i(\vec{r}) = 0.1 \vec{e}_n e^{j \vec{k} \cdot \vec{r}}$$

$$\begin{cases} \vec{e}_i = \cos 60^\circ \vec{e}_z + \sin 60^\circ \vec{e}_x = \frac{\sqrt{3}}{2} \vec{e}_x + \frac{1}{2} \vec{e}_z \\ \vec{e}_n = \frac{\vec{k}_i}{|\vec{k}_i|} = \frac{\sqrt{3}}{2} \vec{e}_x + \frac{1}{2} \vec{e}_z \end{cases}$$

$$|\vec{k}_i| = \omega \sqrt{\mu_0 \epsilon_0} = 2\pi f \sqrt{\mu_0 \epsilon_0} = \frac{2\pi}{3} \text{ rad/m}$$

$$\therefore \vec{k}_i = \frac{2\pi}{3} \left(\frac{\sqrt{3}}{2} \vec{e}_x + \frac{1}{2} \vec{e}_z \right)$$

$$\begin{aligned} \therefore \vec{H}_i(\vec{r}) &= \vec{e}_y 0.1 e^{-j \frac{2\pi}{3} \left(\frac{\sqrt{3}}{2} x + \frac{1}{2} z \right)} \\ &= \vec{e}_y 0.1 e^{-j \frac{\sqrt{3}}{3} \pi x + \frac{\pi}{3} z} \end{aligned}$$

$$\begin{aligned} \vec{E}_i(\vec{r}) &= \vec{z} \vec{H}_i(\vec{r}) \times \vec{e}_n = 120\pi \left(\frac{1}{2} \vec{e}_x - \frac{\sqrt{3}}{2} \vec{e}_z \right) 0.1 e^{-j \frac{\sqrt{3}}{3} \pi x + \frac{\pi}{3} \pi z} \\ &= 12\pi \left(\frac{1}{2} \vec{e}_x - \frac{\sqrt{3}}{2} \vec{e}_z \right) e^{-j \frac{\sqrt{3}}{3} \pi x + \frac{\pi}{3} \pi z} \end{aligned}$$

$$\begin{aligned} \vec{H}_r(\vec{r}) &= -\vec{e}_y 0.1 e^{j \left(\frac{\sqrt{3}}{3} \pi x - \frac{\pi}{3} \pi z \right)} \\ &= -\vec{e}_y 0.1 e^{-j \left(\frac{\sqrt{3}}{3} \pi x - \frac{\pi}{3} \pi z \right)} \end{aligned}$$

$$\begin{aligned} \vec{E}_r(\vec{r}) &= \vec{z} \vec{H}_r(\vec{r}) \times \vec{e}_n', \text{ 其中 } \vec{e}_n' = \frac{\sqrt{3}}{2} \vec{e}_x - \frac{1}{2} \vec{e}_z \\ &= 120\pi \left(\frac{1}{2} \vec{e}_x + \frac{\sqrt{3}}{2} \vec{e}_z \right) e^{-j \left(\frac{\sqrt{3}}{3} \pi x - \frac{\pi}{3} \pi z \right)} \times 0.1 \\ &= 12\pi \left(\frac{1}{2} \vec{e}_x + \frac{\sqrt{3}}{2} \vec{e}_z \right) e^{-j \left(\frac{\sqrt{3}}{3} \pi x - \frac{\pi}{3} \pi z \right)} \end{aligned}$$

(2) 求理想导体表面上的感应电流和电荷密度； $\vec{e}_n$ 是法向

$$\begin{aligned} \vec{e}_n \times (\vec{H}_i - 0) &= \vec{J}_s \quad \text{可知 } \vec{J}_s = \left(\frac{\sqrt{3}}{2} \vec{e}_x + \frac{1}{2} \vec{e}_z \right) \times \left(\vec{e}_y 0.1 e^{-j \frac{\sqrt{3}}{3} \pi x + \frac{\pi}{3} \pi z} \right) \Rightarrow z=0 \\ \vec{e}_n \cdot (\vec{D}_i - 0) &= \rho_s = -\vec{e}_x 0.1 e^{-j \frac{\sqrt{3}}{3} \pi x + \frac{\pi}{3} \pi z} = -\vec{e}_x 0.1 e^{-j \frac{\sqrt{3}}{3} \pi x} \end{aligned}$$

进一步地， $\vec{e}_n \cdot \vec{E}_i = \rho / \epsilon_0$

$$\left(\frac{\sqrt{3}}{2} \vec{e}_x + \frac{1}{2} \vec{e}_z \right) \cdot 12\pi \left(\frac{1}{2} \vec{e}_x - \frac{\sqrt{3}}{2} \vec{e}_z \right) e^{-j \left(\frac{\sqrt{3}}{3} \pi x + \frac{\pi}{3} \pi z \right)} \cdot \epsilon_0 = \rho$$

$$\begin{aligned} \therefore \rho &= 12\pi \epsilon_0 \left(-\frac{\sqrt{3}}{2} \right) e^{-j \left(\frac{\sqrt{3}}{3} \pi x + \frac{\pi}{3} \pi z \right)} \\ &= 6\sqrt{3} \pi \epsilon_0 e^{-j \left(\frac{\sqrt{3}}{3} \pi x + \frac{\pi}{3} \pi z \right)}, \quad z=0 \end{aligned}$$